LaRT v2 User Manual

Contents

1. Introduction **3**

2. Installation and Build **3**

2.1 Prerequisites . . . 3

2.2 Compilation . . . 4

2.3 Running . . . 4

3. Input File Format **4**

4. Grid Configuration **5**

4.1 Cartesian Grid . . . 5

4.2 Grid Geometry . . . 5

4.3 Symmetry and Periodic Boundaries . . . 5

4.4 Distance Units . . . 6

4.5 Grid Type Selection . . . 6

5. Source Configuration **6**

5.1 Source Geometry . . . 6

5.2 Spectral Type . . . 7

5.3 Line Identification . . . 7

5.3.1 Combined Hydrogen + Deuterium Lyα . . . 8

5.3.2 He I 10833 Coherent Triplet Treatment . . . 9

5.3.3 Lyβ Resonance Scattering with Hα + Two-Photon Fluorescence . . . 10

5.3.4 Molecular-Hydrogen (H_2) Effect on Lyα . . . 13

5.4 Frequency and Wavelength Grid . . . 14

6. Medium Properties **15**

6.1 Optical Depth Normalization . . . 15

6.2 Temperature . . . 15

6.3 Density . . . 15

6.4 Velocity Fields . . . 16

7. AMR Grid Mode **17**

7.1 Enabling AMR Mode . . . 17

7.2 Generic AMR Data Format . . . 18

7.3 Extended AMR Physics Parameters . . . 19

7.4 Tau Normalization Convention . . . 19

7.5 Converting RAMSES Data . . . 19
7.6 Converting Illustris/IllustrisTNG Data . . . 20
7.7 Source Geometries in AMR Mode . . . 22

8. Clumpy Medium Mode **22**
8.1 Overview . . . 22
8.2 Enabling Clump Mode . . . 22
8.3 Clump Parameters . . . 23
8.4 Spatially-varying clump properties . . . 24
8.5 Standalone clump generator (`make_clumps.x`) . . . 25
8.6 Overlapping Clumps . . . 26
8.7 Clump Velocity . . . 27

9. Observers and Peel-off **27**
9.1 Observer Placement . . . 28
9.2 Multiple Observers . . . 28
9.3 Image Parameters . . . 28
9.4 Peel-off Output Flags . . . 28
9.5 Sight-line Optical-depth and Column-density Maps . . . 29
9.6 Inside Observer (HEALPix All-Sky) . . . 30

10. Output Files **31**
10.1 Primary Output Arrays . . . 31
10.2 Angular Spectrum `Jmu` . . . 32
10.3 Mean Intensity and Scattering Rate (`CALCJ` / `CALCP` / `CALCPnew`) . . . 32
10.4 Peel-off cube units and conversion to `Jmu` units . . . 34
10.5 Output Control Parameters . . . 35

11. File Formats: FITS and HDF5 **35**
11.1 Format selection . . . 36
11.2 HDF5 schema . . . 36
11.3 Conversion utility . . . 37
11.4 HDF5 library requirements . . . 37

12. Dust and Polarization **37**

13. Parallelism **38**

14. Complete Parameter Reference **38**

15. Python Utilities **43**

15.1 Loading and analysing output: `read_lart.py` 43

15.2 Format-agnostic reader and FITS ↔ HDF5 converter: `lart_io.py` 46

16. Example Input Files **47**

16.1 Uniform Sphere, Lyα, Static 47

16.2 Sphere with Hubble Outflow and Peel-off 47

16.3 Realistic Galaxy (FITS Density Field, Stokes Polarization) 48

16.4 AMR Sphere (Generic Text Format) 48

16.5 Clumpy Sphere with Random Velocities 49

16.6 Illustris/TNG Galaxy (Diffuse Emissivity) 49

1. Introduction

LaRT (**L**yman-**a**lpha **R**adiative **T**ransfer) is a Monte Carlo radiative transfer code for computing the propagation of resonance-line photons through astrophysical neutral-gas media. The default transition is Lyman-α (Lyα, H I 1216 Å), but many other UV/optical/IR lines are supported (see §5.3). Dust scattering and absorption, full Stokes polarization tracking, Hubble-flow and other bulk velocity fields, and spatially varying density, temperature, and velocity from external data files are all supported.

Starting from v2.00, the Cartesian grid is supplemented by:

- **AMR mode** — octree adaptive mesh refinement, allowing spatially varying resolution (§7.).
- **Clumpy medium mode** — a population of compact, optically thick spherical clumps embedded in a transparent inter-clump medium (§8.).

This manual covers v2.00 and v2.10. The only user-visible change in v2.10 is the introduction of `par%grid_type` as a unified selector for the three grid modes; the old `par%use_amr_grid` and `par%use_clump_medium` flags remain valid for backward compatibility.

2. Installation and Build

2.1 Prerequisites

- Fortran 2003+ compiler: Intel `ifort`/`ifx` or GNU `gfortran` ≥ 7.
- CFITSIO library.
- HDF5 library (Fortran bindings), version ≥ 1.14, built with the same compiler used for LaRT. Required for the default build (`HDF5=1`); skip with `HDF5=0` for a FITS-only build.
- MPI library (MPICH or OpenMPI).

2.2 Compilation

```
cd LaRT_v2.00            # or LaRT_v2.10
make                     # produces LaRT.x (HDF5=1 is the default)
make HDF5=0              # FITS-only build (no HDF5 link dependency)
make HDF5_PREFIX=/path   # override HDF5 install location
make DEBUG=1             # full debug flags + runtime checks
make CALCJ=1 CALCP=1     # mean intensity J(x) + scattering rate P_alpha
make F90=mpif90          # override compiler
make clean               # remove .o and .mod files
make cleanall            # also remove executables
make convert_ramses      # auxiliary RAMSES converter
```

`make` recompiles all objects; `make main` recompiles only the sources an edit reaches, following the module dependencies in `src/Makefile.dep` (generated from the `USE` statements), and recompiles everything when a compiler flag changes. `make -j` is safe. `HDF5_PREFIX` defaults to `/data/opt/hdf5_intel_2.2.0` (HDF5 2.2.0) on this machine; set it on the command line or as an environment variable when building elsewhere.

2.3 Running

```
mpirun -np 8 LaRT.x input.in
```

Output is written to `<base>.h5` by default (or `<base>.fits.gz` when `par%file_format = 'fits'` is set), where `<base>` is the input file base name (without `.in`) unless `par%out_file` is set explicitly. See §11. for the two supported file formats and how to choose between them.

3. Input File Format

Input files use Fortran *namelist* syntax. All parameters are prefixed with `par%` and collected in a single `&parameters` block.

```
&parameters
 par%no_photons      = 1e6
 par%temperature     = 1.0e4      ! gas temperature [K]
 par%taumax          = 1.0e4
 par%nx = 101, par%ny = 101, par%nz = 101
 par%rmax            = 1.0
 par%source_geometry = 'point'
 par%spectral_type   = 'voigt'
 par%nxfreq          = 121
/
```

- String values must be enclosed in single quotes.
- Multiple parameters may appear on one line, separated by commas.
- Lines beginning with ! are comments.
- Unspecified parameters retain compiled defaults (listed in §14.).

4. Grid Configuration

4.1 Cartesian Grid

The simulation domain is a rectangular box centered at the origin. By default $x \in [-x_{\max}, +x_{\max}]$ (and similarly for y and z). The cell size is $\Delta x = 2x_{\max}/n_x$.

Parameter	Default	Description
par%nx, par%ny, par%nz	101	Number of cells per axis
par%xmax, par%ymax, par%zmax	1.0	Half-box size per axis (code units)
par%rmax	–999	Spherical boundary: density = 0 for $r > r_{\max}$
par%rmin	–999	Inner cavity: density = 0 for $r < r_{\min}$
par%cone_opening	0.0	Biconical outflow half-opening angle [degrees] along the z-axis. When > 0, density is

If par%input_field or par%dens_file is given, (nx,ny,nz) and (xmax,ymax,zmax) are read from the FITS header automatically.

4.2 Grid Geometry

par%geometry selects the spatial symmetry used for accumulating and saving the mean-intensity array J:

Value	J output geometry
'sphere' (default)	1-D radial profile $J(r)$
'cylinder'	2-D cylindrical $J(\rho, z)$
'rectangle'	3-D Cartesian cube $J(x, y, z)$
'plane_atmosphere'	1-D slab $J(z)$
'spherical_atmosphere'	1-D spherical shell $J(r)$

4.3 Symmetry and Periodic Boundaries

Parameter	Default	Effect
par%xyz_symmetry	.false.	Simulate only the $+x, +y, +z$ octant ($8\times$ RAM saving)
par%xy_symmetry	.false.	Simulate only the $+x, +y$ quadrant ($4\times$ RAM saving)
par%xy_periodic	.false.	Periodic in x, y: photons exiting one face re-enter the opposite

Symmetry options are disabled automatically when peel-off is active.

4.4 Distance Units

Parameter	Default	Description
par%distance_unit	''	Physical unit: 'kpc', 'pc', 'au', or '' (dimensionless)
par%distance2cm	1.0	Alternatively: 1 code unit in cm

In dimensionless mode (par%distance_unit = ''), the gas density is determined entirely by par%taumax and the grid geometry; no physical unit is needed.

4.5 Grid Type Selection

Parameter	Default	Description
par%grid_type†	'car'	'car' (Cartesian), 'amr', or 'clump' (v2.10; v2.00 since 2026-10-20)
par%use_amr_grid	.false.	same as par%grid_type = 'amr'
par%use_clump_medium	.false.	same as par%grid_type = 'clump'

5. Source Configuration

5.1 Source Geometry

par%source_geometry controls the spatial distribution of photon emission:

Value	Description
'point'	Point source at (par%xs_point, par%ys_point, par%zs_point); default $(0,0,0)$
'uniform'	Uniform volume emissivity throughout the grid (or inside par%source_rmax)
'uniform_sphere'	Uniform within a sphere of radius par%source_rmax
'uniform_xy'	Uniform in the $z = 0$ plane
'gaussian'	Gaussian: $\exp(-r^2/2\sigma_r^2)\exp(-z^2/2\sigma_z^2)$; widths par%source_rscale, par%source_zscale
'exponential'	Exponential disc: $\exp(-r/r_s)\exp(-\lvert z\rvert/z_s)$; scales par%source_rscale, par%source_zscale
'exponential_sphere'	Exponential clipped to sphere of radius par%source_rmax
'exponential_cylinder'	Exponential clipped to a cylinder
'sersic' / 'ssh'	Sérsic profile; index par%sersic_m, half-light radius par%Reff
'diffuse_emissivity'	3-D emissivity read from par%emiss_file
'star_file'	Individual point sources from text file par%star_file (columns: x, y, z, L)

Associated parameters:

Parameter	Default	Description
par%xs_point, par%ys_point, par%zs_point	0.0	Point source position (code units)
par%source_rmax	–999	Outer truncation radius for extended sources
par%source_rscale	–999	Radial scale length
par%source_zscale	0.0	Vertical scale height
par%sersic_m	4.0	Sérsic index
par%Reff	1.0	Effective (half-light) radius
par%comoving_source	.true.	Source photons co-moving with the local gas

5.2 Spectral Type

par%spectral_type sets the frequency distribution of emitted photons:

Value	Description
'monochromatic'	Delta function at line center ($x = 0$)
'voigt'	Voigt profile at the local gas temperature
'voigt0'	Voigt profile at the fixed temperature par%temperature0
'gaussian'	Gaussian with σ =par%gaussian_sigma_vel or FWHM=par%gaussian_FWHM_vel [km s^{-1}]
'continuum'	Flat spectrum over [par%velocity_min, par%velocity_max]
'continuum+gaussian'	Fraction $f_{\rm line}$ as Gaussian (FWHM=par%gaussian_FWHM_vel), rest flat continuum; $f_{\rm line} = W_\lambda^v/(W_\lambda^v + \Delta v)$ where $W_\lambda^v = {\rm EW}/\lambda_0\, c$ and Δv is the velocity range. Requires par%EW_line and par%gaussian_FWHM_vel.
'line_prof_file'	Arbitrary profile read from par%line_prof_file

5.3 Line Identification

The atomic transition is selected with par%line_id (default: Lyα).

par%line_id	Transition	Wavelength
(default)	H I Lyα	1215.67 Å
'ly_alpha_HD'	H I + D I Lyα	1215.67 Å (see below)
'ly_beta'	H I Lyβ (+ Hα + 2-photon)	1025.72 Å (see below)
'CII_1334'	C II (+ fluorescence)	1334.53 Å
'CIV_1548'	C IV doublet	1548.19 / 1550.77 Å
'NV_1239'	N V doublet	1238.82 / 1242.80 Å
'OVI_1032'	O VI doublet	1031.91 / 1037.61 Å
'SiIV_1394'	Si IV doublet	1393.76 / 1402.77 Å
'NaI_D'	Na I D doublet	5891.58 / 5897.56 Å
'CaII_HK'	Ca II H & K doublet	3934.78 / 3969.59 Å
'MgII_2796'	Mg II doublet	2796.35 / 2803.53 Å
'AlII_1671'	Al II singlet	1670.79 Å
'SiII_1190' / 'SiII_1193'	Si II doublet (+ fluorescence)	1190.42 / 1193.29 Å
'SiII_1260'	Si II (+ fluorescence)	1260.42 Å
'SiII_1304'	Si II (+ fluorescence)	1304.37 Å
'SiII_1527'	Si II (+ fluorescence)	1526.71 Å
'FeII_2250'	Fe II (+ fluorescence)	2249.88 Å
'FeII_2261'	Fe II (+ fluorescence)	2260.78 Å
'FeII_UV3' / 'FeII_2344'	Fe II UV3 multiplet	2344.21 Å
'FeII_UV2' / 'FeII_2383'	Fe II UV2 multiplet	2382.76 Å
'FeII_UV1' / 'FeII_2600'	Fe II UV1 multiplet	2600.17 Å
'HeI_10833'	He I triplet	10832.06 Å

The numerical line%line_type that controls the dispatch of par%calc_voigt / par%do_resonance (set internally by setup_resonance_line) takes the following values: *1* = singlet (one upward transition); *2* = doublet (two upward transitions sharing a single thermal width, e.g. C IV, N V, O VI, Na I D, Ca II H&K, Mg II, Si IV); *4* = one resonance plus one or more fluorescent decays (most Si II, Fe II UV3); *5* = multiple upward + multiple downward transitions, the most general case (Si II 1190/1193, Fe II UV1, Fe II UV2); *6* = three upward transitions plus one downward transition (He I 10833); *7* = combined H + D Lyα (ly_alpha_HD, two coexisting two-level resonance scatterers, no fine structure); *8* = multiband fluorescence (ly_beta: a single upper level with a resonance branch and a band-transmuting fluorescent branch, see §5.3.3).

5.3.1 Combined Hydrogen + Deuterium Lyα

The line id 'ly_alpha_HD' treats neutral hydrogen and deuterium Lyα as two coexisting two-level resonance scatterers in the same medium. The two transitions sit only 0.33 Å ($\sim$ 82 km/s) apart and have nearly identical atomic constants except for the atomic mass and rest wavelength. The deuterium-to-hydrogen number-abundance ratio is set by

Parameter	Default	Description
`par%D_to_H_ratio`	1.5×10^{-5}	n_D/n_H (cosmic primordial)
`par%include_deuterium`	`.false.`	Convenience flag

Setting `par%include_deuterium = .true.` together with `par%line_id = 'ly_alpha'` promotes the run automatically to `'ly_alpha_HD'`, so existing input files can opt in without changing the line id.
Physical assumptions in this version (v1):

- Both H and D are treated as simple two-level resonance scatterers (`line_type=7`). Fine-structure splitting is *not* included for either species; if `par%fine_structure = .true.` is set together with `'ly_alpha_HD'`, the code emits a warning and disables fine structure.
- The phase function is the simple dipole ($E_1 = 1$, $E_2 = 0$, $E_3 = 1$), identical for both species.
- Oscillator strengths and damping rates are taken to be equal for H and D ($f_D = f_H$, $\Gamma_D = \Gamma_H$ to better than 0.1%). Mass and rest wavelength differences are fully accounted for in the Voigt evaluation and species selection.
- Photon `xfreq` is carried in H-frame Doppler units throughout. D-frame conversions happen on-the-fly in the combined Voigt and scattering routines via constants precomputed at line setup (`line%delta_nu_HD_Hz`, `line%ratio_Dfreq_HD`, `line%ratio_voigta_HD`). No cell-by-cell deuterium arrays are stored.
- During a deuterium scattering event, the perpendicular thermal velocity components and the recoil shift are evaluated using the H atomic constants, which produces a small ($\sim 1/\sqrt{2}$) error in the re-emission shape from D-scattered photons. Because D scatters are rare at cosmic abundance and the dominant D feature (the absorption dip) depends on opacity rather than re-emission shape, this is acceptable for v1. Tracking each photon's species is also not yet implemented.

The Deuterium Lyα rest wavelength used is $\lambda_D = 1215.337$ (NIST vacuum), ~ 0.33 blueward of $\lambda_H = 1215.668$. In LaRT's velocity convention ($v = -c\,\Delta\lambda/\lambda_H$, so negative v is blueshifted), the D Lyα line center sits at $v_D \approx -82\,\mathrm{km/s}$.
A reference example with five validation runs and a Jupyter notebook is provided under `examples/lya_HD/` (see `run.sh` and `inspect_HD.ipynb`). One of those runs is configured for direct comparison against Dijkstra, Haiman & Spaans (2006), ApJ 649, 14, Figure 3 (D/H = 3×10^{-5}, $\tau \sim 10^6$).

5.3.2 He I 10833 Coherent Triplet Treatment

The He I λ10830 multiplet $2^3S_1 \rightarrow 2^3P^o_{0,1,2} \rightarrow 2^3S_1$ is a three-component resonance line (LaRT `line_type=6`). The three upper levels are split by only ~ 0.09 (P_1–P_2, $\sim 2.5\,\mathrm{km/s}$) and ~ 1.25 (P_0–P_2, $\sim 35\,\mathrm{km/s}$), so depending on the gas temperature the components may overlap and interfere.
LaRT offers two treatments of the polarizability parameters (E_1, E_2, E_3) that enter the scattering phase function, selected by the boolean `par%HeI_coherent`:

Toggle	(E_1, E_2, E_3) source
`par%HeI_coherent = .false.` (default, legacy)	Incoherent constants for each component $(0,1,0)$, $(\frac{1}{4},\frac{3}{4},\frac{1}{4})$, $(\frac{7}{20},\frac{13}{20},\frac{3}{4})$ for P_0, P_1, P_2 from Seon `scatter_matrix_v1.1` §9.4. Assigned channel-by-channel based on which upper state the photon was sampled into.
`par%HeI_coherent = .true.`	Frequency-dependent $E_1(\nu), E_3(\nu)$ from the Real-Φ polynomial form of Seon `scatter_matrix_HeI` memo (eqs. 29 & 30, bottom line), with $E_2 = 1 - E_1$.

The polynomial form is, with $D_m \equiv \nu - \nu_{m1}$ for $m = 0, 1, 2$:

$$E_1(\nu) = \frac{3D_2^2D_0^2 + 7D_0^2D_1^2 + 8D_2D_0D_1^2 + 18D_2D_0^2D_1}{4\left[D_2^2D_1^2 + 3D_2^2D_0^2 + 5D_0^2D_1^2\right]}, \tag{1}$$

$$E_3(\nu) = \frac{3D_2^2D_0^2 + 15D_0^2D_1^2 + 8D_2^2D_0D_1 + 10D_2D_0^2D_1}{4\left[D_2^2D_1^2 + 3D_2^2D_0^2 + 5D_0^2D_1^2\right]}, \tag{2}$$

$$E_2(\nu) = 1 - E_1(\nu). \tag{3}$$

The common denominator is a positive-definite sum of squares with the three distinct line centers, so the formula is well-defined at every frequency and recovers the v1.1 §9.4 incoherent values exactly at each line center ($E_1, E_3 = 7/20, 3/4$ at P_2; $1/4, 1/4$ at P_1; $0, 0$ at P_0). It is implemented in `compute_HeI_E_coherent` in `line_mod.f90` (with parallel branches in `scattering_amr.f90` and `line_clump_mod.f90`).

Upper-state selection probability is unchanged. The total cross-section is the incoherent 1:3:5 weighted Lorentzian sum because all J'–J' interference terms cancel in $W_{11} + 2W_{14}$ (the angular integration projects onto the $K = 0$ irreducible component). Both modes therefore sample the upper state with the same probability, so the photon-by-photon frequency redistribution is statistically identical. Differences between the two modes appear only in the phase-function asymmetry E_1 (linear Rayleigh polarizability) and the circular polarizability E_3.

Expected size of the effect. For nebular conditions ($T \sim 10^4$ K) the He-atom Doppler width $\Delta v_D \sim 6.4\,\mathrm{km/s}$ *exceeds* the P_1–P_2 splitting ($\sim 2.5\,\mathrm{km/s}$), so those two components are effectively merged and the inter-line interference is small in the line core. The escape spectrum is essentially unchanged between the two modes. Polarization differences are visible mainly at high τ ($\tau \gtrsim 100$) and in the line wings where photons sample the inter-line frequency range; integrated polarization profiles differ by ~ 5–$15\,\%$ (relative) at $\tau \sim 10^3$. The effect would be larger at lower temperature ($T \lesssim 10^3$ K), where the P_1 and P_2 components are resolved.
A reference test suite is provided under `examples/HeI_coherent_test/` with 20 runs covering {point, uniform-sphere source} × $\{\tau = 0.1, 1, 10, 100, 1000\}$ × {`.false.`, `.true.`} and a comparison notebook `compare.ipynb`.

5.3.3 Lyβ Resonance Scattering with Hα + Two-Photon Fluorescence

Setting `par%line_id = 'ly_beta'` transports H I Lyβ ($\lambda_0 = 1025.7222$ Å) through neutral hydrogen with the fluorescent conversion channel included. Each $1s \to 3p$ absorption branches:

- **88.166 %** back to $1s$: Lyβ resonance re-emission ($A_{3p\to1s} = 1.6725\times10^8\,\mathrm{s}^{-1}$);
- **11.834 %** to $2s$: one Hα photon ($3p\to2s$, 6564.553 Å vacuum, $A_{3p\to2s} = 2.2448\times10^7\,\mathrm{s}^{-1}$), followed by a $2s\to1s$ two-photon continuum ($A = 8.229\,\mathrm{s}^{-1}$; two photons with $\nu_1+\nu_2 = \nu_{\mathrm{Ly}\alpha}$).

The Voigt damping parameter uses the *total* $3p$ width $A_{\mathrm{tot}} = 1.8970\times10^8\,\mathrm{s}^{-1}$, and the downward branching ratios are $[0.88166, 0.11834]$ (`line_type=8`).
The destruction (conversion) probability of 0.11834 per scattering makes Lyβ trapping weak even at large τ_0: a photon survives on average only ~ 8 scatterings before converting, so the problem is the classic ϵ-destruction two-level problem with $\epsilon = 0.11834$. Photon-number bookkeeping per conversion: one Lyβ photon destroyed $\to$ one Hα photon + two two-photon-continuum photons, with the energy check $E(\mathrm{Ly}\beta) = E(\mathrm{H}\alpha)+E(\mathrm{Ly}\alpha)$.

Multiband design. The three emission channels differ in wavelength by factors up to ~ 6.4, so a single `xfreq` grid cannot represent them. LaRT gives each photon a band index (`photon%iband`) and writes independent output spectra:

Band	Channel	Opacity	Transport / output
1	Lyβ (1026 Å)	H I Voigt + dust	Full machinery: `Jin`/`Jout`/`Jabs`, peel-off, `Jmu`, `CALCJ`/`CALCP` on the existing `xfreq` grid
2	Hα (6563 Å)	dust only	Dust scattering (albedo/HG) + absorption; *no* gas opacity ($2s$ metastable/unpopulated)
3	Two-photon (1216 Å–∞)	—	*Analytic* emergent spectrum in $y = \nu/\nu_{\mathrm{Ly}\alpha} \in (0,1)$; no Monte Carlo transport

At a conversion event the packet is transmuted in place (band $1\to2$) with the Hα frequency set by the atom velocity already sampled at the Lyβ absorption, so the physical Ly$\beta\leftrightarrow$Hα frequency correlation is preserved.

Two-photon analytic spectrum. Because band 3 is not transported, the emergent two-photon spectrum is written analytically with zero Monte Carlo variance. Using the Nussbaumer & Schmutz (1984) fit, with $w = y(1-y)$,

$$\frac{dA}{dy} = 202.0\left[w\left(1-(4w)^{0.8}\right)+0.88\,w^{1.53}\,(4w)^{0.8}\right]\,\mathrm{s}^{-1}, \tag{4}$$

whose integral over $(0,1)$ is $16.452\,\mathrm{s}^{-1} = 2\times8.226$ (two photons per decay) and which is symmetric about $y = 1/2$. The emergent spectrum is $J_{2\gamma}(y) = 2W_{\mathrm{conv}}\,P(y)$ with normalized shape $P(y) = (dA/dy)/16.452$ and W_{conv} the total conversion weight.

New input parameters. Dust defaults are the Draine (2003, $R_V = 3.1$) values at the relevant band.

Parameter	Default	Description
par%nxfreq_Ha	0	Hα band spectral bins ($0 \rightarrow$ inherit par%nxfreq)
par%xfreq_max_Ha	0.0	Hα half-range in Doppler units, same $v_{\rm th}$ as band 1 ($0 \rightarrow$ inherit band-1 velocity range)
par%ny_2gam	101	Two-photon y bins over $(0,1)$; 0 disables the tally
par%cext_dust_Ha	3.801e-22	Dust extinction at 6562 Å [cm^2/H] (Draine)
par%albedo_Ha	0.6741	Hα dust single-scattering albedo
par%hgg_Ha	0.4967	Hα Henyey–Greenstein asymmetry g
par%transport_2gamma	.false.	Reserved (dust transport); unimplemented

The derived opacity ratio $R_{\rm H\alpha}$ = cext_dust_Ha/cext_dust = 0.1857 is recomputed at setup (never hardcoded). For par%line_id = 'ly_beta', setup.f90 also re-defaults the band-1 dust constants to the Lyβ (1025.72 Å interpolated) Draine values — cext_dust = 2.047×10^{-21} cm^2/H, par%albedo = 0.2804, par%hgg = 0.6483 (the global defaults are the Lyα row); explicit user overrides still win. All defaults reproduce the prior behavior for every other line: iband stays 1, the new arrays are unallocated, and the new output sections are absent.

Output sections. The following are written to the .h5/.fits.gz output; par%out_merge accumulates each independently.

Section	Length	Contents / attributes
Jout_Ha	nxfreq_Ha	Emergent Hα spectrum; attrs nxfreq_Ha, Dxfreq_Ha, Xfreq1/2_Ha, Dwave_Ha, lambda0_Ha = 6564.553 Å, and the conservation weights W_esc1/W_abs1/W_conv/W_esc2/W_abs2
Jabs_Ha	nxfreq_Ha	Hα dust-absorbed spectrum (only when dust absorption is present)
J2gam	ny_2gam	Two-photon spectrum; attrs ny_2gam, dy_2gam, Wconv_pp; y centers at $(i-\frac{1}{2})dy$
Pconv_*	(grid)	CALCP builds only. The Lyβ $\rightarrow$Hα conversion rate per atom in each cell or bin (the Hα source function). The shape follows the grid geometry: Pconv_1D (radial), Pconv_2D (cylindrical), Pconv_3D (per cell), or Pconv_AMR (per leaf), with the same axes and normalization as the matching Pa section
peel_Ha	(nxfreq_Ha, nxim, nyim)	Hα peel-off cube per observer (in the peel-off file)

Core-skip is disabled automatically. `setup.f90` forces `par%core_skip = .false.` for `ly_beta` and prints a `NOTE`. Each skipped core scattering would have carried an 11.8 % conversion chance, so core-skip would systematically underproduce Hα and overestimate Lyβ escape. Acceleration is unnecessary anyway, since the 0.118 destruction probability already caps the scattering count at ~ 8.

`CALCJ` / `CALCP` behavior and the `Pconv` map. As for any resonance line, a `CALCJ` build accumulates the in-medium mean intensity $J(\nu, \cdot)$ and a `CALCP` build the scattering rate per atom P_α. For `ly_beta` these are computed for band 1 (the resonant Lyβ radiation) only; the dust-only band 2 and the analytic band 3 carry no in-medium J.
A `CALCP` build additionally writes a `Pconv` map — the local *conversion* rate per atom, i.e. the rate at which Lyβ excitations are converted into the Hα + two-photon channel ($3p \to 2s$) rather than re-emitted as Lyβ ($3p \to 1s$). It is accumulated exactly like P_α (the same storage in each cell or bin, position binning, and volume-weighted normalization), but is tallied *only at conversion events* instead of at every scattering. `Pconv` is therefore the emission source function for the fluorescent products: the local Hα (and two-photon) emissivity is directly proportional to it. Because every scattering converts with the fixed branching probability $A(3p \to 2s)/A_{\rm tot}(3p) = 0.11834$, the two maps satisfy, cell by cell and up to Monte Carlo noise,

$$\frac{P_{\rm conv}(\mathbf{r})}{P_\alpha(\mathbf{r})} \longrightarrow 0.11834, \tag{5}$$

which doubles as an internal consistency check. The section is named `Pconv_1D/2D/3D/Pconv_AMR` following the grid geometry, mirroring the `Pa` sections, and `read_lart.py` loads it as `Pconv1/Pconv2/Pconv3D/Pconv_leaf`.

Conservation bookkeeping. LaRT prints a weight budget,

$$N_{\rm esc}({\rm Ly}\beta) + N_{\rm conv} + N_{\rm dustabs}({\rm Ly}\beta) = N_{\rm photons}, \tag{6}$$

$$N_{\rm esc}({\rm H}\alpha) + N_{\rm dustabs}({\rm H}\alpha) = N_{\rm conv}, \tag{7}$$

which map to the `W_esc1/W_abs1/W_conv/W_esc2/W_abs2` attributes stored in `Jout_Ha`.

Support matrix. Both the Cartesian and AMR grids are supported. The following are guarded off for `ly_beta` and abort or fall back with a clear message: the clumpy medium, the HEALPix inside-observer path, Stokes polarization, fine structure (`par%fine_structure = .true.`), non-trivial symmetry folding, and the slab (`par%xy_periodic`) geometry. Two-photon transport (`par%transport_2gamma`) is reserved but unimplemented (analytic tally only), and deuterium Lyβ is not modeled.

5.3.4 Molecular-Hydrogen (H_2) Effect on Lyα

Several H_2 Lyman (B – X) and Werner (C – X) band lines lie within a few hundred km s^{-1} of H Lyα. A Lyα photon whose local-frame frequency coincides with one of these lines can be absorbed into an excited electronic state of H_2, which then either (a) decays back to the pumping level and re-emits near Lyα (resonance scattering off H_2, with probability $p_{\rm scat} = A_{ul}/A_{\rm tot}$), or (b) fluoresces to another wavelength and is lost from the Lyα field (Neufeld 1990, ApJ 350, 216; Black & van Dishoeck 1987). Neufeld's conversion probability $\simeq 1$ means the destruction branch dominates ($p_{\rm scat} \sim 0.08$–0.09), but the return branch is physically included. This is enabled for `par%line_id = 'ly_alpha'` by

setting `par%h2_model` $\neq$ `'none'`; it works on the Cartesian *and* AMR grids, and peel-off sight-lines are attenuated by H_2. The default `par%h2_model = 'none'` skips the whole H_2 path (bit-identical to runs without H_2).

The `'neufeld'` model treats the two B – X lines closest to Lyα: R(6) at +14.1 km s^{-1} [B(1,7) ←X(2,6)] and P(5) at +99.2 km s^{-1} [B(1,4) ←X(2,5)]. Their line-centre opacities are $s_i = f_{\rm H2}\, f(v,J)(f_{\rm osc,i}/f_{\rm osc,Ly\alpha})$ relative to the H I Lyα opacity, with LTE level populations $f(v,J)$ computed from a partition function (ortho J-odd weight 3, para J-even weight 1) at the H_2 temperature. The H_2 temperature `par%h2_temperature` is an *independent* input (not the gas temperature) that sets both the LTE populations and the H_2 Doppler width.

Parameter	Default	Meaning
`par%h2_model`	`'none'`	`'none'` | `'neufeld'` | `'lte'`
`par%f_H2`	0.0	molecular fraction $n(H_2)/n(HI)$
`par%h2_temperature`	1000.0	H_2 temperature [K] (LTE pops. + Doppler width)
`par%h2_pure_absorption`	`.false.`	`.true.` $\Rightarrow p_{\rm scat} = 0$ (Neufeld limit)
`par%h2_hi_width`	`.false.`	benchmark: H_2 line at the H I Doppler width
`par%h2_data_dir`	`''`	directory holding `energy_X.dat`

At the end of a run LaRT prints the absorbed-fraction budget ($1-f_e = W_{\rm H2abs}/N_{\rm phot}$, the resonance-scatter count, and the pumping weights for each line) and stores it as Spectrum-header keywords (`H2ABS`, `H2SCAT`, `H2PUMP1..2`); `read_lart` exposes it as `h2_budget`. The implementation reproduces Neufeld's analytic slab absorbed fraction (Fig. 20) to $< 1\%$ at $N(HI) \gtrsim 10^{20}$ cm^{-2} and the P(5)/R(6) pumping ratio (Fig. 21); Cartesian and AMR agree to $\sim$ 1–2%. Validation and a parameter-sweep spectrum notebook are in `examples/h2_test/`.

5.4 Frequency and Wavelength Grid

Internally, photon frequencies are tracked in the dimensionless Doppler variable:

$$x = \frac{\nu - \nu_0}{\Delta\nu_D}, \qquad \Delta\nu_D = \frac{\nu_0}{c}\, b_D, \qquad b_D = \sqrt{\frac{2kT}{m} + b_{\rm turb}^2}, \tag{8}$$

where ν_0 is the line-center frequency, T is the local gas temperature, m is the atomic/ionic mass, and $b_{\rm turb}$ = `par%bturb` is a spatially uniform turbulent velocity added *in quadrature* to the local thermal width (`par%bturb` ≤ 0, the default, recovers the pure thermal width). The same b_D, evaluated with the *local* cell temperature, sets the line opacity, the Voigt parameter, the scattering redistribution, and the velocity-to-Doppler conversion identically on the Cartesian, AMR, and clump grids. Output spectra are binned onto a uniform grid in x by default. A wavelength or velocity grid can be requested instead:

Parameter	Default	Description
par%nxfreq	121	Number of spectral bins
par%xfreq_min, par%xfreq_max	auto	Limits of x-grid
par%xfreq0	0.0	Frequency offset applied to all emitted photons
par%nwavelength	0	Number of wavelength bins (overrides x-grid)
par%wavelength_min, par%wavelength_max	—	[Å]
par%nvelocity	0	Number of velocity bins (overrides x-grid)
par%velocity_min, par%velocity_max	—	[km s^{-1}]

The range is set automatically from the mean medium temperature when no limits are provided.

6. Medium Properties

6.1 Optical Depth Normalization

The overall opacity is scaled by one of the following (specify at most one):

Parameter	Description
par%taumax	Line-center τ_0 from the grid center to the $+z$ boundary along the z-axis
par%tauhomo	Same, assuming a uniform density field (scales an inhomogeneous field proportionally)
par%N_HImax	H I column density from center to $+z$ boundary [cm^{-2}]
par%N_HIhomo	Same, assuming uniform density

For AMR grids, par%taumax is the pole-to-face optical depth (box center $\rightarrow +z$ face along z), consistent with the Cartesian convention for a sphere of radius $R = L/2$.

6.2 Temperature

Parameter	Default	Description
par%temperature	10^4 K	Uniform gas temperature
par%temperature0	$= T$	Temperature for spectral_type='voigt0' source
par%bturb	0.0	Turbulent velocity dispersion added in quadrature to thermal width [km s^{-1}]
par%temp_file	''	3-D temperature cube [K]; FITS or HDF5

6.3 Density

Analytic profiles. Without external files, the density is set analytically via the opacity scale parameters (par%taumax etc.) combined with an optional radial profile:

Profile type	Key parameter	Functional form
Uniform (sphere or box)	`par%rmax`	Constant $n_{\rm H}$ inside $r < r_{\rm max}$
Spherical shell	`par%rmin`, `par%rmax`	Constant $n_{\rm H}$ between $r_{\rm min}$ and $r_{\rm max}$
Exponential	`par%density_rscale`	$n \propto \exp(-r/r_s)$
Power law	`par%density_alpha`	$n(r) \propto (r_{\rm max}/r)^{\alpha_D}$; α_D=0: uniform, α_D=2: isothermal

External gridded data (FITS or HDF5).

The reader (`read_grid_data.f90`) detects the format from the filename extension (`.fits[.gz]`, `.h5`, or `.hdf5`) and uses the same code path for both. HDF5 inputs expect the array at `/<section>/data` (any section name); shape information is read from the dataset itself, so FITS-style `NAXIS` keywords are not required.

Parameter	Description
`par%dens_file`	3-D H I number density cube [cm^{-3}]
`par%temp_file`	3-D temperature cube [K]
`par%velo_file`	4-D velocity cube (v_x, v_y, v_z) [km s^{-1}]
`par%emiss_file`	3-D emissivity cube [photons s^{-1} cm^{-3}]
`par%input_field`	Base name: reads `<base>.dens.fits.gz`, `.temp.`, `.velo.` (multi-file convention; FITS-only)

If `par%taumax` (or `par%N_HImax`) is specified alongside a density file, the density is rescaled so that the target optical depth is achieved.

6.4 Velocity Fields

`par%velocity_type` selects the bulk velocity model. The default (empty string) is a static medium.

Value	Key parameters	Velocity law
`'hubble'`	`par%Vexp`	$\boldsymbol{v}(r) = V_{\rm exp}(r/r_{\rm max})\hat{r}$; $V_{\rm exp} > 0$: outflow
`'constant_radial'`	`par%Vexp`	$\boldsymbol{v}(r) = V_{\rm exp}\hat{r}$ (constant magnitude)
`'power_law'`	`par%Vexp`, `par%velocity_alpha`	$\boldsymbol{v}(r) = V_{\rm exp}(r/r_{\rm max})^{\alpha_V}\hat{r}$
`'linear_decelerate'`	`par%Vexp`	$\boldsymbol{v}(r) = V_{\rm exp}\frac{r_{\rm max}-r}{r_{\rm max}-r_{\rm min}}\hat{r}$
`'parallel_velocity'`	`par%Vx`, `par%Vy`, `par%Vz`	Uniform $\boldsymbol{v} = (V_x, V_y, V_z)$ [km s^{-1}]
`'ssh'`	`par%Vpeak`, `par%rpeak`, `par%DeltaV`	Bilinear profile (Song, Seon & Hwang 2020)
`'rotating_solid_body'`	`par%Vrot`	$\boldsymbol{v} = V_{\rm rot}(-y, x, 0)/r_{\rm max}$
`'rotating_galaxy_halo'`	`par%Vrot`, `par%rinner`	$v_\phi = V_{\rm rot}\, r_{\rm cyl}/r_{\rm inner}$ for $r_{\rm cyl} < r_{\rm inner}$, $V_{\rm rot}$ outside (Kym et al. 2026)

The `'ssh'` profile rises linearly to $V_{\rm peak}$ at $r_{\rm peak}$, then shifts by ΔV toward $r_{\rm max}$:

$$v(r) = \begin{cases} V_{\rm peak}\, r / r_{\rm peak} & r < r_{\rm peak} \\ \left[V_{\rm peak} + \Delta V \, \frac{r - r_{\rm peak}}{r_{\rm max} - r_{\rm peak}} \right] \hat{r} & r \geq r_{\rm peak}. \end{cases} \tag{9}$$

A velocity FITS file (`par%velo_file`) may be combined with a systematic `par%velocity_type`.

Other velocity-related parameters:

Parameter	Default	Description
`par%Omega`	28.0	Shear rate $[(\mathrm{km\,s^{-1}})\,\mathrm{kpc^{-1}}]$ for TIGRESS data
`par%recoil`	`.false.`	Include photon-recoil frequency shift in scattering
`par%core_skip`	`.false.`	Enable resonance-core skipping acceleration
`par%core_skip_global`	`.false.`	With `core_skip`: use the old volume-averaged $x_{\rm crit}$ instead of the cell-by-cell formula (bench

Core-skip details. When `core_skip = .true.` and `core_skip_global = .false.` (the default combination), LaRT recomputes the core-skip threshold $x_{\rm crit}$ at every scattering using the local cell:

$$a\tau_{\rm cell} = a_{\rm V}(\text{cell})\,\texttt{rhokap}(\text{cell})\,\Delta\ell_{\rm face}, \qquad x_{\rm crit} = \begin{cases} (a\tau_{\rm cell})^{1/3}/5 & a\tau_{\rm cell} > 1 \\ 0 & \text{otherwise} \end{cases}$$

where $\Delta\ell_{\rm face}$ is the photon's distance to the nearest cell boundary. This is Smith et al. (2015, Eq. 35) applied per cell, following the RASCAS convention. Inhomogeneous boxes such as a dense galactic core in a sparse hot halo require this cell-by-cell treatment; the `core_skip_global = .true.` branch uses a single volume-averaged $x_{\rm crit}$ that collapses to zero whenever $\langle a\tau \rangle < 1$ and is retained only for comparison with the older LaRT behavior.

7. AMR Grid Mode

7.1 Enabling AMR Mode

v2.00:

```
par%use_amr_grid = .true.
par%amr_file     = 'grid.h5'
```

v2.10 (preferred):

```
par%grid_type = 'amr'
par%amr_file  = 'grid.h5'
```

LaRT reads only the *generic AMR format* (text, FITS, or HDF5). Simulation data from RAMSES or other codes must first be converted using the standalone converter (`convert_ramses_to_generic.x` or the Python equivalent). The parameter `par%amr_type` defaults to `'generic'`; setting it to `'ramses'` is no longer supported and produces an error directing the user to the converter.

Parameter	Default	Description
par%amr_type	'generic'	Only 'generic' is supported
par%amr_file	''	Path to generic AMR file (.h5, .fits.gz, .dat)

7.2 Generic AMR Data Format

The generic AMR format stores leaf-cell data in text, FITS binary table, or HDF5. Nine columns are mandatory; six optional columns provide pre-computed physics quantities that LaRT can use directly.

Mandatory columns (9):

Column	Unit	Description
x, y, z	code units	Leaf cell center coordinates
level	—	AMR refinement level
nH (or gasDen, dens)	cm^{-3}	Total hydrogen number density
T	K	Gas temperature
vx, vy, vz	$\mathrm{km\,s}^{-1}$	Gas velocity

Optional columns (detected by name in HDF5/FITS):

Column	Unit	Description
metallicity	mass fraction	Raw Z from simulation
xHI	dimensionless	Neutral hydrogen fraction
n_e	cm^{-3}	Electron density
n_ion	cm^{-3}	Scattering ion density (metal lines)
emissivity	$\mathrm{cm}^{-3}\,\mathrm{s}^{-1}$	Line emissivity rate (Lyα, C IV, or O VI)
ndust	cm^{-3}	Dust pseudo-number density

When an optional column is present in the file, LaRT uses it directly. When absent, LaRT computes the quantity internally using the model selected by the extended physics parameters (see below).

Plain text format: the header gives the number of leaf cells and the box length — either as a single legacy line "<nleaf> <boxlen>" or as NLEAF / BOXLEN keyword lines in any order. Lines starting with # (and blank lines) are ignored anywhere in the file. Every data line lists one leaf cell with exactly the nine mandatory columns; the optional physics columns are not supported in text format (use FITS or HDF5 for those).

```
# either:  8192  100.0        (legacy one-line header)
BOXLEN  100.0
NLEAF   8192
# x  y  z  level  nH  T  vx  vy  vz
-49.5  -49.5  -49.5  1  1.0e-3  1.0e4  0.0  0.0  0.0
...
```

Text-format coordinates must be given in the *centered* frame, $[-L/2, +L/2]$: the text format carries no origin information, so the reader always assumes a centered box. (FITS and HDF5 files instead carry `ORIGINX/Y/Z` headers and may use either convention.)

7.3 Extended AMR Physics Parameters

These parameters control how LaRT computes physical quantities when the corresponding optional column is absent from the generic file. All defaults reproduce the existing behavior.

Parameter	Default	Description
`par%ionization_model`	`'cie_formula'`	Neutral fraction model: `'cie_formula'` (single-formula CIE), `'cie_table'` (Gnat & Sternberg 2007), `'full_neutral'` ($x_{\rm HI} = 1$), `'from_file'` (require `xHI` column)
`par%dust_model`	`'global_dgr'`	Dust model: `'global_dgr'` (use `par%DGR`), `'laursen09'` (metallicity-based, Laursen et al. 2009), `'from_file'` (require `ndust` column)
`par%emissivity_model`	`'none'`	Lyα emissivity: `'none'`, `'caseB'` (Case B recombination + collisional excitation), `'from_file'`
`par%ion_model`	`'none'`	Metal-line ion density: `'none'`, `'solar_cie'` (solar abundances $\times$ CIE ion fractions), `'from_file'`
`par%metallicity_global`	-1	Global metallicity (Z, mass fraction). Used by `'laursen09'` and `'solar_cie'` when no metallicity column is present.
`par%Z_ref`	0.0134	Reference solar metallicity (Asplund et al. 2009)
`par%f_ion_dust`	0.01	Ionized-gas dust fraction (Laursen et al. 2009)

7.4 Tau Normalization Convention

For AMR, `par%taumax` is the line-center optical depth from the box center to the $+z$ face along the z-axis. This matches the Cartesian sphere convention where `par%taumax` is the center-to-surface optical depth, provided $r_{\rm max} = L/2$.

7.5 Converting RAMSES Data

RAMSES binary snapshots must be converted to the generic format before use with LaRT. Two converters are provided:

Fortran:

```
make convert_ramses
convert_ramses_to_generic.x /path/to/ramses 91 output.h5 kpc
convert_ramses_to_generic.x /path/to/ramses 91 output.h5 kpc \
    --compute-physics --metallicity 0.0134
```

Python:

```
python convert_ramses_to_generic.py /path/to/ramses 91 -o output.h5 \
    --compute-physics --metallicity 0.0134
```

The `--compute-physics` flag pre-computes `xHI`, `n_e`, and `emissivity` using the CIE formula. If `--metallicity` is given, `ndust` is also computed via the Laursen et al. (2009) model.

7.6 Converting Illustris/IllustrisTNG Data

Illustris and IllustrisTNG simulations use the AREPO moving-mesh code, which stores gas data on an unstructured Voronoi tessellation. Since LaRT requires an octree AMR grid, the Voronoi data must be resampled onto an adaptive octree. The converter `convert_illustris_to_generic.py` handles this automatically:

```
python convert_illustris_to_generic.py cutout.hdf5 -o galaxy.h5 \
    --output-unit kpc --level-max 8 --compute-physics \
    --ionization from_illustris --boxsize 200
```

The converter reads `PartType0` fields from an HDF5 snapshot or API cutout, converts Illustris comoving+h code units to physical (kpc, cm^{-3}, K, km s^{-1}), builds an adaptive octree via density and velocity gradient refinement, and assigns leaf-cell properties from the Voronoi mesh. By default this uses a nearest-neighbor KD-tree lookup (physically exact for a Voronoi tessellation); `--resample-method gaussian` instead applies a volume-weighted Gaussian kernel smoothing with an adaptive, smoothing length proportional to the local cell size, for a continuous (but softened) field. Key options:

Flag	Description
`--grid-type`	Output grid: `amr` (adaptive octree, default) or `cartesian` (uniform N^3)
`--ngrid`	Cells per side for `--grid-type cartesian` (default 128)
`--level-max`	Maximum octree level (default 8; AMR mode only)
`--dens-threshold`	Density gradient threshold (default 0.3; AMR mode only)
`--match-resolution`	Also refine to match local Voronoi cell size (AMR mode only)
`--resample-method`	`nearest` (Voronoi-exact, default) or `gaussian` (volume-weighted kernel smoothing)
`--kernel-size-factor`	Gaussian mode: adaptive smoothing length $h_i = f \times r_{\mathrm{eff},i}$ (default 1.0)
`--kernel-size`	Gaussian mode: fixed smoothing length, overriding the adaptive factor
`--compute-physics`	Compute `xHI`, `n_e`, emissivity, `ndust`
`--emissivity-line`	`lya` (Case B, default), `civ` (CIE C IV), `ovi` (CIE O VI)
`--ionization`	`from_illustris` (TNG native), `cie`, or `full_neutral`
`--sfr-treatment`	`cap-temperature` (default), `exclude`, or `as-is`
`--api-key`	Download cutout from TNG public API
`--plot`	Diagnostic slice plot (area-filled by default; `--plot-scatter` for cell-center scatter)

The `--sfr-treatment cap-temperature` option (default) caps the temperature of star-forming cells at 2×10^4 K, since their `InternalEnergy` reflects an effective equation of state, not the actual ISM temperature.

Cutouts can be downloaded directly from the TNG public API (`tng-project.org`; free registration required):

```
python convert_illustris_to_generic.py \
    --api-key YOUR_KEY --simulation TNG50-1 --snap 99 \
    --subhalo-id 0 -o galaxy.h5 --compute-physics
```

When the converted grid file contains an `emissivity` column, `source_geometry = 'diffuse_emissivity'` will automatically use it as the photon source distribution without requiring `par%emiss_file` to be set. Alternatively, set `par%emissivity_model = 'caseB'` to compute the emissivity on the fly from n_{H}, T, and x_{HI}.

Cartesian grid output. In addition to the adaptive octree, the converter can produce a uniform $N \times N \times N$ Cartesian grid:

```
python convert_illustris_to_generic.py cutout.hdf5 \
    -o galaxy_cart.h5 --grid-type cartesian --ngrid 128 \
    --output-unit kpc --compute-physics --ionization cie
```

The output file stores all quantities as 3D image sections (one per physical variable), read by LaRT via `par%cart_file`:

```
par%cart_file     = 'galaxy_cart.h5'
```

```
par%distance_unit = 'kpc'
```

This uses the standard Cartesian raytrace with no AMR overhead. Optional columns (`xHI`, `emissivity`, `ndust`) are applied automatically when present.
For full details of the Illustris data format, unit conversions, and the Voronoi-to-octree algorithm, see `docs/Illustris_data_structur`

7.7 Source Geometries in AMR Mode

In v2.00, AMR supports `'point'` and `'uniform'`. In v2.10, most Cartesian geometries (Gaussian, exponential, Sérsic, `star_file`, etc.) are also available; refer to §5.1 and the v2.10 release notes.

8. Clumpy Medium Mode

8.1 Overview

Clumpy medium mode places N spherical clumps of radius r_c inside a spherical outer domain of radius R. The inter-clump medium is vacuum. All opacity resides inside the clumps. By default, clump positions are drawn by Random Sequential Adsorption (RSA) so that clumps never overlap. Optionally, overlapping clumps can be generated or loaded from an external FITS file; LaRT detects overlaps automatically after placement and engages an event-based multi-component raytrace in that case (see §8.6).

8.2 Enabling Clump Mode

v2.00:

```
par%use_clump_medium = .true.
```

v2.10 (preferred):

```
par%grid_type = 'clump'
```

8.3 Clump Parameters

Parameter	Default	Description
`par%rmax`	1.0	Outer sphere radius (code units)
`par%rmin`	–999	Inner cavity radius (the default sentinel is treated as 0). Clumps are placed only in the shell $r_{\min} \le r \le r_{\max}$
`par%clump_radius`	—	Individual clump radius (same units)
`par%clump_tau0`	—	Line-center τ_0 from clump center to its surface
`par%clump_f_cov`	—	Covering factor $C_f = N r_c^2/R^2$
`par%clump_f_vol`	—	Volume filling factor $f_V = N(r_c/R)^3$
`par%clump_N_clumps`	—	Total number of clumps N
`par%temperature`	10^4 K	Clump gas temperature
`par%clump_sigma_v`	0.0	Gaussian σ of random bulk clump velocities [km s^{-1}]
`par%save_clump_info`	`.false.`	Write clump positions and velocities to a FITS binary table
`par%clump_input_file`	`''`	If non-empty, load the clump population from this FITS file (see §8.5) instead of generating it internally
`par%cone_opening`	0.0	Biconical outflow half-opening angle [degrees]. When > 0, clumps are placed only inside the bicone (both hemispheres). With `par%clump_fully_inside`, clumps are additionally rejected if they protrude outside the cone boundary.
`par%clump_fully_inside`	`.true.`	If `.true.` (default), reject any candidate whose outermost edge would protrude past the outer sphere ($d_{\rm center}+r_c > R$) or into the inner cavity ($d_{\rm center}-r_c < r_{\min}$); if `.false.`, restore the legacy behavior in which only the center is required to lie in $[r_{\min}, R]$
`par%clump_allow_overlap`	`.false.`	If `.true.`, skip RSA overlap rejection during internal generation: clumps are placed at uniformly random positions. After placement `check_has_overlap()` is called automatically; if any overlapping pair is found the overlap-aware raytrace is engaged. See §8.6

Specify exactly one of `par%clump_f_cov`, `par%clump_f_vol`, or `par%clump_N_clumps`.

For the opacity of each clump, specify *one* of `par%clump_tau0`, `par%clump_NHI`, or `par%clump_nH`. As a convenience, when none of those three is set the system-level radial-sightline targets `par%taumax` and `par%N_HImax` are also accepted for procedurally generated clumps: LaRT back-solves the peak opacity of each clump so the realised radial sightline from the box centre to $r = R$ matches the requested `par%taumax` (or `par%N_HImax`). In the uniform-clump case the back-solve uses the closed-form $\tau_{\max} = \frac{4}{3} f_{\rm cov}\, \tau_{c,0}$ (and similarly for $N_{\rm HI,max}$); in profile mode it uses the radial quadra-

ture derived from par%clump_radius_profile, par%clump_density_profile, and par%clump_number_profile. par%N_gasmax is treated as an alias for par%N_HImax in clump mode (HI-only line). When a pre-built clump population is loaded via par%clump_input_file, any one of par%taumax, par%tauhomo, par%N_HImax (=par%N_gasmax), or par%N_HIhomo (=par%N_gashomo) may be supplied as the rescale target. Every cl_rhokap(i) is multiplied by a single factor α so the chosen target is exactly reproduced; the other three scalars are derived consistently from the rescaled distribution. Priority order when several targets are given: par%taumax > par%tauhomo > par%N_gasmax > par%N_gashomo. See the *Clumpy Medium Algorithm Description*, §"System-level fallback".

When par%rmin is set, par%clump_f_vol and par%clump_f_cov are interpreted relative to the shell volume $V_{\rm shell} = \frac{4}{3}\pi(R^3 - r_{\rm min}^3)$ and the shell sightline factor $R^2 + R\,r_{\rm min} + r_{\rm min}^2$. With par%rmin unset (default –999, treated as 0) these reduce exactly to the original $(R/r_c)^3$ and $(R/r_c)^2$ formulas, preserving prior runs. When par%clump_fully_inside is on, the realized filling factors fall slightly below the requested target whenever r_c is a non-trivial fraction of $R - r_{\rm min}$, since the available placement volume shrinks from $\frac{4}{3}\pi(R^3 - r_{\rm min}^3)$ to $\frac{4}{3}\pi\left[(R - r_c)^3 - (r_{\rm min} + r_c)^3\right]$. Specify a slightly larger target if exact matching is required. See the *Clumpy Medium Algorithm Description* document, "Fully-Inside Constraint" subsection, for the placement details.

8.4 Spatially-varying clump properties

Three optional radial-profile inputs let each clump's radius, internal H I density, and spatial number density vary with distance from the box center. Built-in shapes are 'constant' (default), 'powerlaw', 'gaussian', 'exponential', and 'file' (5-column ASCII table). Setting any flag to a value other than 'constant' switches par%init_clumps to the profile-aware path.

Parameter	Default	Description
par%clump_radius_profile	'constant'	shape of $r_c(r)$
par%clump_radius_alpha	0.0	power-law index for $r_c(r)$
par%clump_radius_r0	0.0	power-law scale for $r_c(r)$; $\leq 0 \Rightarrow r_{\rm max}$
par%clump_density_profile	'constant'	shape of internal $n_H(r)$
par%clump_density_alpha	0.0	power-law index for $n_H(r)$
par%clump_density_r0	0.0	power-law scale for $n_H(r)$; $\leq 0 \Rightarrow r_{\rm max}$
par%clump_number_profile	'constant'	shape of spatial $n_{cl}(r)$
par%clump_number_alpha	0.0	power-law index for $n_{cl}(r)$
par%clump_number_r0	0.0	power-law scale for $n_{cl}(r)$; $\leq 0 \Rightarrow r_{\rm max}$
par%clump_profile_file	''	5-column tabulated profile (used when any axis = 'file')

For non-uniform profiles the closed-form $f_{\rm cov}$ / $f_{\rm vol}$ / N relation does not hold. Instead the profile-aware quadrature back-solves the spatial normalization so that whichever of par%clump_f_cov, par%clump_f_vol, or par%clump_N_clumps is provided is matched, and the other two are reported on stdout. See the *Clumpy Medium Algorithm Description* document for the integral definitions and the implementation outline.

8.5 Standalone clump generator (`make_clumps.x`)

A self-contained Fortran driver, `make_clumps.x`, builds a clump population without running the photon transport. It is no longer linked against MPI or the LaRT physics stack; the only run-time dependency is the FITS/HDF5 output facade. All parameters are passed as command-line flags (one flag per `par%clump_*` field plus the relevant global parameters):

```
make make_clumps                                          # builds make_clumps.x
./make_clumps.x --rmax 1.0 --clump_radius 1e-3 \
                --clump_f_cov 5 --clump_tau0 1e3 \
                --temperature 1e4 --line_id ly_alpha \
                --seed 12345 -o run_clumps.fits.gz
```

Run `./make_clumps.x --help` for the full flag list. The output filename's extension chooses the container: `.fits` / `.fits.gz` produce a FITS binary table, `.h5` / `.hdf5` produce an HDF5 table. In both cases the header keywords and column schema match the clump table that LaRT itself writes with `par%save_clump_info`, so the file is directly consumable by LaRT through `par%clump_input_file`.

A pure-Python equivalent, `python/make_clumps.py`, ships in the same repository; the two drivers expose essentially the same flag set and write the same schema. All sampling paths are supported in both: uniform RSA, radial profiles (`powerlaw`, `gaussian`, `exponential`), biconical geometry (`--cone_opening`, including the `clump_fully_inside` angular-subtension test), every opacity-input mode (`--clump_tau0` / `--clump_NHI` / `--clump_nH` / `--taumax` / `--N_HImax`), and the same systematic velocity families (`hubble`, `constant_radial`, `power_law`, `linear_decelerate`, `parallel_velocity`, `ssh`, `rotating_solid_body`, `rotating_galaxy_halo`) on top of the Gaussian `par%clump_sigma_v` component.

```
python python/make_clumps.py --rmax 1.0 --clump_radius 1e-3 \
       --clump_f_cov 5 --clump_tau0 1e3 --temperature 1e4 \
       --line_id ly_alpha --seed 12345 -o run_clumps.fits.gz
```

Both drivers are deterministic in `--seed`: re-running with the same flag set and the same seed reproduces the table bit-for-bit. The two drivers use independent RNG implementations, however, so for a given seed the Fortran and Python outputs are statistically equivalent but not byte-identical; choose whichever fits the host environment. Either output is dropped into a LaRT run by setting `par%clump_input_file`.

To consume the file, set `par%clump_input_file` on the LaRT input:

```
&parameters
 par%grid_type        = 'clump'
 par%rmax             = 1.0
 par%clump_radius     = 0.01
 par%clump_tau0       = 1.0e3
 par%temperature      = 1.0e4
 par%no_photons       = 1e6
 par%clump_input_file = 'clumps_clumps.fits.gz'
```

```
/
```

When par%clump_input_file is non-empty, init_clumps skips the profile-detection and RSA-generation paths entirely and reads the population from the file. read_clumps_fits accepts files with any subset of the clump columns R_CLUMP/RHOKAP/TEMP; missing columns are filled from the corresponding header keyword (CL_RAD/RHOKAP/TEMP_CL). For the RHOKAP slot the column names DENSITY and DENS are also accepted as aliases, in priority order RHOKAP $\rightarrow$ DENSITY $\rightarrow$ DENS. The interpretation depends on which name was used: RHOKAP (and the header-keyword fallback) is taken as line-centre opacity per code unit and stored directly into κ_i, while DENSITY or DENS are taken as the neutral-hydrogen number density $n_{HI,i}$ [cm^{-3}] and converted to opacity via $\kappa_i = n_{HI,i} \cdot$ line%cross0$/\Delta\nu_{D,i} \cdot$ par%distance2cm. The conversion requires par%distance_unit (and therefore par%distance2cm) to be set; if not, the read path emits a warning and falls back to treating the values as opacity.

The clump file also carries DISTUNIT (par%distance_unit) and DIST_CM (par%distance2cm) keywords written by LaRT itself. On read, these are adopted into par%distance_unit / par%distance2cm when the user did NOT supply them in the input namelist; the criterion for "user-supplied" is par%distance_unit /= '' *and* par%distance2cm > 1.0, so the file values take effect only when the namelist leaves both at their defaults. This lets a clump file generated with par%distance_unit = 'kpc' be replayed in a downstream run without having to repeat that setting, and gives a meaningful par%distance2cm to the DENSITY/DENS opacity conversion above. External pipelines that write LaRT-compatible HDF5 clump files must store DISTUNIT as a fixed-length ASCII string (h5py: string_dtype(encoding='ascii', length=N)); variable-length strings are detected on read and silently skipped (a warning is printed).

If any of par%taumax, par%tauhomo, par%N_HImax (=par%N_gasmax), or par%N_HIhomo (=par%N_gashomo) is specified together with par%clump_input_file, every loaded cl_rhokap(i) is multiplied by a single factor α = target/realised so the chosen system-level scalar matches the input target; the other three scalars are then derived consistently from the rescaled population. Priority among multiple targets: par%taumax $>$ par%tauhomo $>$ par%N_gasmax $>$ par%N_gashomo. If none of the four is set, the loaded opacities are taken as-is and the four scalars are reported as computed from the file. See *Clumpy Medium Algorithm Description*, §"System-level fallback" and §"Derived Global Optical-Depth Quantities" for the formulas.

8.6 Overlapping Clumps

By default LaRT guarantees non-overlapping clumps through RSA placement and all raytrace paths assume this property. Two situations can produce overlapping populations:

1. A clump population loaded from an external FITS file via par%clump_input_file may contain overlapping clumps.

2. Setting par%clump_allow_overlap = .true. skips the RSA rejection step so that internally generated clumps are placed at uniformly random positions and will typically overlap whenever the clumps are large relative to the domain.

After the CSR acceleration grid is built, `check_has_overlap()` scans all clump pairs. If any overlapping pair is found it prints

```
Overlap check: overlapping clumps detected -- using overlap-aware raytrace.
```

sets an internal flag `has_overlap = .true.`, and routes all subsequent raytrace calls to the overlap-aware variants automatically. No input change is required; the detection and dispatch happen at run time.

Physical model for overlapping clumps

Two clumps sharing a volume with different bulk velocities $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are treated as two independent gas components coexisting at the same spatial location. The total line opacity at global-frame frequency x is

$$\kappa_{\text{total}}(x) = \sum_i \rho \kappa_i H(x - u_{\text{los},i}, a_i), \qquad u_{\text{los},i} = \frac{\boldsymbol{v}_i \cdot \hat{k}}{\Delta\nu_{D,i}}. \tag{10}$$

When a scattering event occurs in an overlap region the interacting atom belongs to one clump, chosen by opacity-weighted sampling $P(\text{clump } i) = \kappa_i / \kappa_{\text{total}}$.

Quick-start example

Large clumps with low covering factor guarantee many overlaps while keeping the run fast:

```
par%grid_type           = 'clump'
par%clump_radius        = 0.3
par%clump_f_cov         = 2.0
par%clump_allow_overlap = .true.
par%no_photons          = 1e5
par%spectral_type       = 'voigt'
```

Reference inputs: `clump_overlap_mono.in`, `clump_overlap_sigv.in`, `clump_overlap_peel.in` in `examples/clump_sphere/`.

For a full algorithmic description see the *Clumpy Medium Algorithm Description* document, Section 15 “Overlap-Aware Raytrace”.

8.7 Clump Velocity

A systematic flow can be added on top of the random component (`par%clump_sigma_v`) via `par%velocity_type`. Supported values in clump mode: `'hubble'`, `'constant_radial'`, `'power_law'`, `'linear_decelerate'`, `'parallel_velocity'`, `'ssh'`, `'rotating_solid_body'`, `'rotating_galaxy_halo'`.

9. Observers and Peel-off

The peel-off (peeling-off) technique shoots a photon toward the observer at every scattering event, allowing spectral images to be constructed without noise. It is activated by setting `par%nxim` and `par%nyim` to non-zero values.

9.1 Observer Placement

Direction vector:

```
par%obsx = 0.0, par%obsy = 0.0, par%obsz = 1.0  ! look down +z axis
par%distance = 100.0                             ! observer-center separation
```

Spherical angles:

```
par%alpha = 45.0   ! azimuthal angle [deg]
par%beta  = 30.0   ! elevation angle [deg]
par%distance = 100.0
```

`par%gamma` sets the rotation of the detector plane about the line of sight. If `par%distance` is omitted it defaults to $100 \times r_{\max}$ (or the maximum box half-extent).

9.2 Multiple Observers

Multiple observers are specified as arrays:

```
par%alpha = 0.0  45.0  90.0  135.0
par%beta  = 0.0   0.0   0.0    0.0
```

Up to 99 simultaneous observers are supported.

9.3 Image Parameters

Parameter	Default	Description
`par%nxim`, `par%nyim`	0	Image pixel counts; peel-off enabled when > 0
`par%dxim`, `par%dyim`	auto	Pixel angular size [deg] (defaults to cover the full grid)

When `par%dxim`/`par%dyim` are not set, the pixel size is chosen to cover the full sphere:

$$\delta\theta = \frac{1}{N_x/2} \arcsin\left(\frac{r_{\max}}{d}\right). \tag{11}$$

9.4 Peel-off Output Flags

Parameter	Default	Description
`par%save_peeloff`	`.false.`	Save 1-D peel-off spectra per observer
`par%save_peeloff_2D`	`.false.`	Save peel-off spectral image cubes (x-y-ν)
`par%save_peeloff_3D`	`.false.`	Save full 3-D peel-off data
`par%save_sightline_tau`	`.false.`	Compute deterministic sight-line maps from each observer pixel: gas $\tau(\nu)$, $N_{\rm HI}$, and dust τ (when `par%DGR`>0). See §9.5.

9.5 Sight-line Optical-depth and Column-density Maps

When par%save_sightline_tau = .true., LaRT integrates the opacity along each observer pixel's line of sight and writes a separate FITS file <base>_tau<obs>.fits.gz for every observer. The file contains:

HDU	Content
TAU_gas	gas optical depth $\tau(\nu, x, y)$
N_gas	gas column density $N_{\rm HI}(x, y)$
TAU_dust	dust optical depth $\tau_{\rm dust}(x, y)$ (only when par%DGR> 0)

The pixel grid uses the same WCS as the peel-off image (par%nxim, par%nyim, par%dxim, par%dyim); the frequency axis matches the spectrum output (par%nxfreq, par%velocity_min, par%velocity_max). All three grid modes (Cartesian, AMR, clump) are supported with the appropriate raytrace dispatch.

Standalone driver make_sightline_tau.x

The same sight-line integration is exposed as a standalone executable that skips the photon-transport stage entirely:

```
make sightline_tau                              # builds make_sightline_tau.x
mpirun -np N ./make_sightline_tau.x sl.in       # writes <base>_tau.fits.gz
```

sl.in is an ordinary LaRT input file; the driver internally sets par%save_peeloff and the sight-line flag to .true. and otherwise honors every namelist parameter. Useful for parameter sweeps where only the deterministic maps are needed (no Monte-Carlo transport). Unlike make_clumps.x, multiple MPI ranks do speed this driver up: pixels are divided across par%nproc via loop_divide and reduced with reduce_mem.

The folder examples/sightline_tau/ contains a reference set of six runs that exercise every grid mode and observer configuration the driver supports:

File	Grid mode	Observer
car_sightline.in	Cartesian	outside (rectangular FITS)
amr_sightline.in	AMR	outside (rectangular FITS)
clump_sightline.in	clumpy sphere	outside (rectangular FITS)
car_inside.in	Cartesian	inside (HEALPix all-sky)
amr_inside.in	AMR	inside (HEALPix all-sky)

The shipped run.sh executes all six in sequence; make_amr_sphere_data.py regenerates the AMR data file from a uniform-sphere physics model. The companion notebook plot_sightline_tau.ipynb displays the column-density maps, selected $\tau(x, y, v)$ slices, the all-sky HEALPix mollview maps and the sky-mean line profile.

tau_huge cap (sight-line vs. photon transport). The raytrace routines used by the peel-off (Cartesian, AMR and clump) early-exit when the running optical depth exceeds $\tau_{\rm huge} = 300$ (tau_huge in define.f90); beyond it $\exp(-\tau) < 5 \times 10^{-131}$, so accumulating further cannot change the peel-off weight. Until 2026-10-20 the cap was

745.2, the double-precision underflow point of exp(-τ). The sight-line variant `raytrace_to_edge_tau_gas_amr` exists precisely to *report* the integrated τ, however large, and therefore runs without that cap (matching the Cartesian `raytrace_to_edge_tau_gas` in `raytrace_car.f90`). The same applies to `raytrace_to_dist_tau_gas_amr` used by the inside-observer HEALPix path.

9.6 Inside Observer (HEALPix All-Sky)

Setting `par%nside` > 0 switches the entire output and peel-off chain to the *inside-observer* path: the observer is placed at a chosen position (o_x, o_y, o_z) *inside* the medium, and each escaping photon is binned into an all-sky HEALPix map by the direction from the observer back to the photon. This mode is intended for "galactic" observations – e.g., placing the observer at the Sun's position inside a model of the Galactic ISM, or inside a simulated galaxy.

Activation:

```
par%nside = 8                    ! HEALPix Nside parameter; total npix = 12*nside^2
par%obsx  = 0.3                  ! observer position (code units)
par%obsy  = 0.0
par%obsz  = 0.0
par%save_peeloff    = .true.
par%save_peeloff_2D = .true.     ! also save the 2D map (sum over frequency)
```

When `par%nside` > 0, `setup.f90` automatically sets `par%observer_located_inside = .true.` and switches:

- the observer arrays from rectangular images to HEALPix all-sky maps with $N_{\rm pix} = 12\,{\rm Nside}^2$;
- the peel-off routines to the `_inside` variants (which use `vec2pix` to bin the peeled weight by observer-pointing direction);
- `output_reduce`, `output_normalize` and `write_output` to the `_inside` variants.

In v2.10 the dispatch is performed by the type-bound methods of `grid_base_type` (`grid_class.f90`); the `grid_car_type` and `grid_amr_type` concrete types each branch on `par%observer_located_inside` and forward to the appropriate `_inside` or `_outside` concrete routine.

Multiple observers. As with the rectangular path, up to 99 observer positions are supported by passing arrays:

```
par%nside = 8
par%obsx  = 0.0  0.3  0.0
par%obsy  = 0.0  0.0  0.5
par%obsz  = 0.0  0.0  0.0
```

Output files. In inside-observer mode the `_obs.fits.gz` file holds 2D HEALPix arrays of shape `(npix, nxfreq)` with extensions `Direct` (unscattered direct contribution) and `Scattered` (scattered contribution). When `par%save_peeloff_2D = .true.` a separate `_obs2D.fits.gz` file is written with shape `(npix,)` holding the same arrays summed over frequency. The HEALPix maps use the RING ordering convention. The `Spectrum` HDU (`Jout`, `Jin`, `Jabs`) is always written regardless of observer mode.

Source scope. The inside-observer path supports the same internal sources as the standard path – `point`, `uniform`, `gaussian`, `exponential`, `sersic`, `diffuse_emissivity` – but *not* the external-illumination geometries (`point_illumination`, `stellar_illumination`, `plane_illumination`), which are only physically meaningful for an observer outside the medium. Stokes polarization is not yet implemented for the inside-observer path.

Grid-mode support. The HEALPix inside-observer path is available for both Cartesian (`par%grid_type = 'car'`) and AMR (`par%grid_type = 'amr'`) grids. In AMR mode, dedicated `_inside_amr` peeling, output and sight-line routines are used (see the *LaRT AMR Description* document for implementation details). Clumpy medium (`par%grid_type = 'clump'`) is not currently supported with the inside-observer path.

Sightline-tau output. When the inside-observer path is active, an additional `_sightline.fits.gz` file is produced that records τ and N(HI) from the observer position to each of the six box faces ($\pm x$, $\pm y$, $\pm z$). This is useful for cross-checking the optical depth structure seen by the observer.

Smoke test. The `examples/amr_sphere_generic/` directory ships `sphere_amr_inside_test.in` (AMR uniform sphere) and `sphere_car_inside_test.in` (Cartesian 64^3 counterpart), both with `par%nside = 8` and an off-center observer at $(o_x, o_y, o_z) = (0.3, 0, 0)$. These produce equivalent HEALPix maps within the documented AMR octree volume-averaging tolerance.

10. Output Files

Outputs are written either as gzip-compressed FITS files (`.fits.gz`) or as HDF5 files (`.h5`). HDF5 is the default; set `par%file_format = 'fits'` in the namelist to switch. Both formats share the same logical layout (one section per FITS HDU $\leftrightarrow$ one HDF5 group), so the conversion is loss-less and runs in both directions via `python/lart_io.py` (see §11. for the schema and converter, §15. for the Python helper).

10.1 Primary Output Arrays

Array	FITS extension	Description
`Jout`	`JOUT`	Emergent spectrum integrated over all escape directions
`Jin`	`JIN`	Injected (intrinsic) source spectrum
`Jabs`	`JABS`	Spectrum of photons absorbed by dust
`Jmu`	`JMU`	Emergent spectrum binned by $\mu = \cos\theta_z$

Jout is the main science output. It is normalized per unit frequency interval and per unit area of the bounding sphere. In dimensionless units (no `par%distance_unit`):

$$J_{\rm out}(x) = \frac{\text{escaped photon weight in bin } x}{n_{\rm ph}\,\Delta x\,2\pi\,\pi R^2}, \tag{12}$$

where $R = r_{\rm max}$ (Cartesian) or $R = L_{\rm box}/2$ (AMR), and $\Delta x = 1$ in units of the Doppler width.

Jin requires `par%save_Jin = .true.`

Jabs requires `par%save_Jabs = .true.`

Jmu requires `par%save_Jmu = .true.`; see §10.2.

The total flux is conserved: $\int J_{\rm out}\,dx + \int J_{\rm abs}\,dx = \int J_{\rm in}\,dx$ (in the absence of other losses).

10.2 Angular Spectrum `Jmu`

Setting `par%save_Jmu = .true.` adds a 2-D image extension `EXTNAME='Jmu'` of shape (n_x, n_μ) that records the emergent spectrum binned by the z-component of the escape direction, $\mu \equiv \cos\theta_z = \hat{k}_z$. This is useful for

- verifying isotropy of homogeneous models (every μ bin should match `Jout` within Poisson noise),
- characterizing the angular dependence of escape in non-isotropic models (for example, a slab or rotating halo).

Binning.

- `par%nmu` bins of width $\Delta\mu$ tile the μ range uniformly.
- Default `par%nmu = 11` (odd) places $\mu = 0$ at a bin center, so the perpendicular-escape spectrum is preserved as a single channel. Even `par%nmu` places $\mu = 0$ at a bin edge and cleanly separates the $+z$ and $-z$ hemispheres.
- Range:
  - `par%xyz_symmetry = .true.` folds escapes onto the $+z$ hemisphere, so $\mu \in [0, +1]$ and $\Delta\mu = 1/n_\mu$.
  - Otherwise $\mu \in [-1, +1]$ and $\Delta\mu = 2/n_\mu$.

  The header keywords `nmu`, `mu_min`, `dmu` record these values, and full WCS keywords (`CTYPE1='XFREQ'`, `CTYPE2='MU'`, `CRVAL*`, `CDELT*`) are written to the image header.

Normalization. `Jmu` uses the same denominator as `Jout` multiplied by n_μ, so that

$$\frac{1}{n_\mu}\sum_{i=1}^{n_\mu} J(x;\mu_i) = J_{\rm out}(x) \tag{13}$$

holds exactly. Equivalently, in a homogeneous, isotropic medium each μ bin reproduces `Jout` up to Poisson fluctuations, providing a cheap regression test for new geometries.

Modes. Available with all three grid modes: Cartesian, AMR octree, and the clumpy medium.

10.3 Mean Intensity and Scattering Rate (`CALCJ` / `CALCP` / `CALCPnew`)

Building with

```
make CALCJ=1 CALCP=1 CALCPnew=1     # any subset may be enabled
```

adds accumulators for the mean-intensity spectrum $J(x)$ inside the medium (`CALCJ`) and for the scattering rate per atom P_α, counted either per resonance-scattering event (`CALCP`) or with the faster path-length estimator of Seon & Kim (2020) (`CALCPnew`). These accumulators work on **both** the Cartesian grid and the AMR octree grid; the clumpy medium is not yet supported.

Geometry selection (`par%geometry_JPa`). The storage shape is controlled by `par%geometry_JPa`: 3 = full 3-D (per cell / per leaf), 2 = cylindrical (r, z) bins, 1 = spherical radial bins, -1 = plane-parallel z bins. If unset, the code picks radial bins when `par%rmax` > 0 (or 3-D when `par%save_all` is set), z bins for `par%xy_periodic` slabs, and 3-D otherwise. **AMR note:** for radial (1) or cylindrical (2) binning on the AMR grid, `par%rmax` > 0 must be set explicitly in the input file (e.g. `par%rmax` = $L_{\rm box}/2$); otherwise the automatic selection falls back to the leaf-by-leaf 3-D shape, because the AMR box size is not yet known when the selection is made. The number of radial bins defaults to $2^{\ell_{\rm max}-1}$ (capped at 512) and can be overridden with `par%nr`.

Output sections.

`par%geometry_JPa`	Cartesian	AMR octree
3	`Jx_3D` $(n_x, n_{\rm x}, n_{\rm y}, n_{\rm z})$, `Pa_3D`	`Jx_AMR` $(n_x, n_{\rm leaf})$, `Pa_AMR` $(n_{\rm leaf})$
2	`Jx_2D`, `Pa_2D`	`Jx_2D`, `Pa_2D`
1, -1	`Jx_1D`, `Pa_1D`	`Jx_1D`, `Pa_1D`

`CALCPnew` sections carry the suffix `_new` (`Pa_1D_new`, `Pa_AMR_new`, ...), so `CALCP` and `CALCPnew` can be enabled together without a section-name collision. The AMR sections also carry the bin-axis keywords `geom_JPa` and `nr` / `rmax` / `dr` (radial) or `nz` / `zmin` / `dz` (z bins), so readers can reconstruct the bin grid without the input AMR file; `read_lart.py` does this automatically. The leaf arrays are ordered by the octree leaf index; the leaf positions (c_x, c_y, c_z, c_h) follow from the input AMR data file (read it with `AMR_grid.py` to paint maps). `par%out_merge` accumulation works for all of these sections.

Binning and normalization. J is the volume-averaged mean intensity per unit frequency in each bin; P_α is the number of scatterings per atom per photon. On the AMR grid the binning is *position-based*: each path segment deposits into the bin containing its midpoint (each scattering event into the bin containing its position), independent of the local leaf size, and the normalization divides by the gas volume overlapping the bin (computed at setup by sub-sampling every leaf with up to 8^3 points; the plane overlap is analytic). Profiles therefore remain correct even where the leaves are much larger than the bins — e.g. the coarsely refined core of a refined-at-boundary octree — at the cost of a small sub-sampling error in the bin volumes ($\lesssim$ a few percent for coarse leaves). The leaf-indexed shape (3) stores each leaf separately and divides by the leaf volume.

Memory. `Jx_AMR` is a `real64` array of shape $(n_{\rm xfreq}, n_{\rm leaf})$ *per MPI rank*: at $n_{\rm xfreq} = 121$ and $n_{\rm leaf} = 10^6$ this is $\sim$1 GB per rank. The startup log reports the allocated size and warns above 1 GB; prefer `par%geometry_JPa` = 1 for spherically symmetric problems.

Validation. For a uniform sphere ($\tau_0 = 100$, $T = 10^4$ K, central point source), the AMR radial profiles agree with a Cartesian 64^3 run to 1–2% (rms) in both $J(x, r)$ and $P_\alpha(r)$ at matched resolution, and to $\sim$10% per bin on a refined-at-boundary octree whose core leaves are $8\times$ wider than the bins; for a uniform slab (`par%xy_periodic`) the plane profiles agree to 0.6% (rms). The `CALCP` and `CALCPnew` estimators agree with each other to Monte-Carlo noise.

10.4 Peel-off cube units and conversion to `Jmu` units

When peel-off is enabled (`par%nxim`, `par%nyim` > 0 and `par%save_peeloff_3D = .true.`), each observer's `<base>_obs_NNN.fits.gz` file contains `Scattered` and `Direct` 3-D image cubes, plus optionally `Direct0`. These are written by `output_sum_rect.f90` as a *specific intensity per pixel*, normalized so that

$$I_{\rm peel}(x,y,\nu) \;=\; \frac{\text{photon weight in pixel } (x,y) \text{ and bin } \nu}{n_{\rm ph}\;\delta\Omega_{\rm pix}\;\Delta\nu\;d_{\rm cm}^2}, \tag{14}$$

where $\delta\Omega_{\rm pix} = (\texttt{CD2_2}\cdot\texttt{CD3_3})(\pi/180)^2$ is the solid angle of each pixel, $\Delta\nu$ is `Dxfreq` or `Dwave` (set by `par%intensity_unit`), $d_{\rm cm}^2$ is the square of the distance unit in cm, and $n_{\rm ph}$ is the photon count.

Conversion to `Jmu` units. `Jout` and `Jmu` are normalized by $4\pi R^2$ (the source surface area) and the outward solid angle 2π, whereas the peel-off cube is normalized by the solid angle of each pixel and $d_{\rm cm}^2$. Summing the cube over pixels and rescaling brings the two onto a common axis:

$$J_{\rm peel}(\nu) \;=\; \Big(\sum_{(x,y)} I_{\rm peel}(x,y,\nu)\Big)\cdot\frac{4\pi D^2\,\delta\Omega_{\rm pix}}{A_{\rm surface}\cdot 2\pi}, \tag{15}$$

where D is `DISTANCE` (in code units) and $A_{\rm surface}$ is the source surface area in code units, with the geometry-dependent forms mirroring `output_sum_rect.f90`:

Geometry	$A_{\rm surface}$ (code units)	Notes
sphere	$4\pi r_{\rm max}^2$	default; clumpy medium also uses this
box (rectangle)	$8\,(x_{\rm max}\,y_{\rm max}+y_{\rm max}\,z_{\rm max}+z_{\rm max}\,x_{\rm max})$	full surface of $[-x_{\rm max},x_{\rm max}]\times\ldots$ box
slab (`par%xy_periodic`)	$2\,(2x_{\rm max})(2y_{\rm max})$	two unbounded boundaries
AMR octree	4π	equivalent to $r_{\rm max}=1$ (cf. §7.)

The bin width $\Delta\nu$ cancels because both `Jmu` and `I_peel` are already normalized per bin. At convergence and for an axisymmetric source,

$$J_{\rm peel}(\nu;\alpha,\beta) \;\to\; J_{\rm mu}(\nu;\mu=\cos\beta). \tag{16}$$

The Python helper `LaRTOutput.plot_peel_jmu_compare()` (in `python/read_lart.py`) implements Eq. (15) with the surface area chosen automatically from `par%geometry`, `par%rmax`, and the spectrum-header `XMAX/YMAX/ZMAX/XY_PER`. See also `examples/sphere/plot_Jpeel_conversion.ipynb` and `examples/clump_sphere/verify_peel_jmu.ipynb`.

Component decomposition (clumpy medium). For a point source inside a clumpy sphere with covering factor $f_{\rm cov}$, the fraction of photons that escapes without scattering is $\exp(-f_{\rm cov})$ (Poisson statistics), and the fraction that scatters at least once is $1-\exp(-f_{\rm cov})$. Jmu tallies *both* populations. The peel-off cube splits them:

- `Scattered`. The MC estimator over many scatter events averages efficiently, so

$$\int J_{\rm peel}^{\rm scatt}\,d\nu \;\approx\; \left(1-e^{-f_{\rm cov}}\right)\int J_{\rm mu}\,d\nu, \tag{17}$$

  and the equality holds across all observer angles within Monte-Carlo noise.

- `Direct`. For a *point* source, every photon shares the same line of sight to a given observer, so the direct contribution is the binary realization of $\exp(-\tau_{\rm LOS})$: either ~ 1 (clear LOS) or ~ 0 (blocked by an intervening clump) per observer. Jmu, by contrast, is azimuth averaged and captures direct escapes uniformly. Adding the binary `Direct` and the smooth `Scattered` therefore only matches `Jmu` *in expectation* – it can deviate substantially per observer at finite photon counts.

For shape comparisons in clumpy runs, prefer the scattered component (`component='scatt'` in `plot_peel_jmu_compare()`) so that Eq. (17) provides a clean, deterministic relation.

10.5 Output Control Parameters

Parameter	Default	Description
`par%file_format`	`'hdf5'`	Output container: `'hdf5'` (→ `.h5`) or `'fits'` (→ `.fits.gz`); see §11.
`par%out_file`	`''`	Output file name (defaults to `<input>.h5`; extension is corrected to match `par%file_format` with a warning); derived files are written next to it
`par%out_merge`	`.false.`	Accumulate into existing output instead of overwriting
`par%save_backup`	`.false.`	On merge, copy the prior output to `<base>_NN.{h5,fits.gz}` before overwriting
`par%out_bitpix`	0	Storage precision: 0 = auto (dynamic-range driven), −32 = float32, −64 = float64
`par%save_Jin`	`.false.`	Save injected spectrum
`par%save_Jabs`	`.false.`	Save dust-absorbed spectrum
`par%save_Jmu`	`.false.`	Save angular spectrum $J(x;\mu)$ (§10.2)
`par%nmu`	11	Number of μ bins for `Jmu`
`par%save_all`	`.false.`	Save full 3-D $J(\nu,x,y,z)$ array
`par%save_direc0`	`.false.`	Save photon escape direction distributions
`par%save_radial_profile`	`.false.`	Save 1-D radial J profile
`par%save_sightline_tau`	`.false.`	Save optical depth along sightlines
`par%intensity_unit`	`''`	Physical unit label written to the FITS header

`par%out_merge = .true.` is useful for noise reduction: run the same setup multiple times and merge the results. All input parameters must be identical across runs.

11. File Formats: FITS and HDF5

LaRT writes one logical layout that lives in either FITS or HDF5 files. The choice is made by `par%file_format` (default `'hdf5'`); the filename extension is corrected to match if the user-supplied `par%out_file` disagrees.

11.1 Format selection

- `par%file_format = 'hdf5'` (default) writes `.h5` files via the `hdf5io_mod` backend. Build with `HDF5=1` (the Makefile default).
- `par%file_format = 'fits'` writes `.fits.gz` files via the `fitsio_mod` backend (CFITSIO). Works regardless of the HDF5 build option.

The mismatch warning is issued once on rank 0, e.g. if `par%out_file = 'foo.fits.gz'` is given alongside `par%file_format = 'hdf5'`:

```
WARNING: par%out_file (foo.fits.gz) does not match
         par%file_format = hdf5
         Overriding par%out_file to: foo.h5
         (set par%file_format explicitly to suppress this warning.)
```

All derived filenames (peel-off `_obs`, sight-line `_tau`, `_allph`, backup) follow `par%file_format` as well, so the main output and its derived files always share a format.

11.2 HDF5 schema

Each FITS HDU maps to one HDF5 group; the HDU's header keywords become attributes on that group; the HDU's data is either a single `/<name>/data` dataset (image-style) or one dataset per `BinTable` column (table-style). Example for a sphere run with peel-off:

```
run.h5
  /Spectrum                                        (group, table-style)
    /Spectrum/Xfreq          (1-D dataset, float32)
    /Spectrum/velocity       (1-D dataset, float32)
    /Spectrum/wavelength     (1-D dataset, float64)
    /Spectrum/Jout           (1-D dataset, float32)
    /Spectrum/Jin            (1-D dataset, float32)
    /Spectrum/@nphotons      (attribute)
    /Spectrum/@taumax        (attribute)
    /Spectrum/@Nsc_gas       (attribute)
    ...

run_obs.h5
  /Scattered                                       (group, image-style)
    /Scattered/data          (3-D dataset, chunked + gzip)
    /Scattered/@CD1_1        (WCS attribute)
    ...
  /Direct
    /Direct/data
```

```
    /Direct/@EXTNAME       = 'Direct'


run_tau.h5
  /TAU_gas/data, /N_gas/data, [/TAU_dust/data when DGR > 0]
```

Sections appear in their write order on both the root group and inside each subgroup (creation-order tracking is enabled), so h5py iteration returns columns and HDUs in the same order LaRT wrote them. Datasets larger than 4096 elements are chunked and compressed with gzip(4); smaller ones are stored contiguous. The on-disk storage type follows par%out_bitpix (auto / float32 / float64) just as for FITS.

11.3 Conversion utility

The Python helper python/lart_io.py reads either format and can convert between them losslessly:

```
python lart_io.py info  run.h5
python lart_io.py convert run.fits.gz run.h5
python lart_io.py convert run.h5      run.fits.gz
```

The round-trip preserves all data bit-identically; only string keyword names are uppercased on the FITS leg, matching the FITS card-name convention.

11.4 HDF5 library requirements

LaRT links against the HDF5 Fortran library (libhdf5_fortran.a plus the C core libhdf5.a, version $\geq$ 1.14). The Fortran .mod files are compiler-specific: an HDF5 build linked against gfortran cannot be used with Intel ifx or ifort, and vice versa. Set par%HDF5_PREFIX on the make command line to choose a build that matches the LaRT compiler.

12. Dust and Polarization

Parameter	Default	Description
par%DGR	0.0	Dust-to-gas ratio relative to the Milky Way value
par%albedo	0.32536	Single-scattering albedo ω
par%hgg	0.67617	Henyey–Greenstein asymmetry parameter g
par%use_stokes	.false.	Enable full Stokes polarization RT
par%scatt_mat_file	''	Mueller matrix file for polarized dust scattering
par%use_reduced_wgt	.false.	Reduce photon weight by albedo instead of destroying absorbed photons

When par%DGR = 0 (default), there is no dust and all photon weight is conserved. Non-zero par%DGR adds dust extinction and Henyey–Greenstein scattering. When par%use_stokes = .true., a Mueller matrix file must be provided; files for the Weingartner–Draine Milky Way model are included in the data/ directory.

13. Parallelism

Parameter	Default	Description
par%no_photons	10^6	Total photon packets (floating-point, converted to int64)
par%use_master_slave	.true.	Dynamic load balancing via master–worker algorithm
par%num_send_at_once	100	Photons dispatched to each worker per batch
par%nprint	10^5	Progress-report interval [photons]
par%iseed	0	Mersenne Twister seed; 0: each rank reads one from /dev/urandom (fallback: clock and pro
par%launch_sequence	'random'	'sobol': quasi-random launch and first flight (below)
par%qmc_seed	0	Sobol scramble seed; 0: drawn by rank 0 from /dev/urandom

With par%launch_sequence = 'sobol' the launch of a photon packet and its first flight are drawn from an Owen-scrambled Sobol point (src/qmc_mod.f90) addressed by the global photon number, so the launch set does not depend on the number of MPI ranks; every later draw stays on the Mersenne Twister. The dimensions are: 1–3 the source frequency (continuum position, Box–Muller Gaussian, Cauchy part of a Voigt source, or the line/continuum choice of 'continuum+gaussian'), 4 the upward component of a multiplet, 5–6 the isotropic direction, 7–9 the position of uniform sphere, cylinder, box and plane sources, and 10 the first-flight optical depth of the forced first scattering. Alias-table and rejection draws stay on the Mersenne Twister. Each run is one randomized replicate; errors are estimated from runs with different par%qmc_seed, and successive par%out_merge runs need different seeds (the default par%qmc_seed = 0 draws a new one). For a Si II 1193 point source with a continuum spectrum ($\tau = 1$, 2^{16} photons, 8 replicates) the variance of Jout fell by a factor 745, of the direct peel-off spectrum by 1761 and of the total scattered peel-off flux by 8.4, while the scattered peel-off spectrum, set by the atomic velocities drawn at scattering, changed by 1.4; for Lyα at $\tau = 10^4$ no change was measurable.

With par%use_master_slave = .true., one MPI rank acts as dispatcher and the rest as workers. This balances workloads when scattering rates vary across photon packets. With .false., each rank processes an equal share of par%no_photons.

14. Complete Parameter Reference

Table 1 lists commonly used parameters. Parameters marked † are new or changed in v2.10.

Table 1: LaRT parameter reference.

Parameter	**Default**	**Description**
Simulation control		
par%no_photons	10^6	Number of photon packets
par%nprint	10^5	Progress interval [photons]

continued on next page

(continued)

Parameter	Default	Description
par%iseed	0	Mersenne Twister seed
par%launch_sequence	'random'	'random' or 'sobol'
par%qmc_seed	0	Sobol scramble seed
Cartesian grid		
par%nx, par%ny, par%nz	101	Cells per axis
par%xmax, par%ymax, par%zmax	1.0	Half-box size (code units)
par%rmax	–999	Spherical boundary radius
par%rmin	–999	Inner cavity radius
par%geometry	'sphere'	J output geometry (§4.2)
par%xyz_symmetry	.false.	$+x,+y,+z$ octant only
par%xy_symmetry	.false.	$+x,+y$ quadrant only
par%xy_periodic	.false.	Periodic in x,y
par%distance_unit	''	Physical unit ('kpc', 'pc', 'au')
par%distance2cm	1.0	1 code unit in cm
Grid mode		
par%grid_type†	'car'	'car', 'amr', 'clump'
par%use_amr_grid	.false.	same as grid_type='amr'
par%use_clump_medium	.false.	same as grid_type='clump'
Source		
par%source_geometry	'point'	See §5.1
par%xs_point, par%ys_point, par%zs_point	0.0	Point source position
par%source_rmax	–999	Source truncation radius
par%source_rscale	–999	Radial scale length
par%source_zscale	0.0	Vertical scale height
par%sersic_m	4.0	Sérsic index
par%Reff	1.0	Effective radius
par%comoving_source	.true.	Source co-moving with local gas
par%emiss_file	''	Emissivity FITS cube
par%star_file	''	Point-star list (x,y,z,L)
Line and spectrum		
par%line_id	Lyα	Atomic transition (§5.3)

continued on next page

(continued)

Parameter	Default	Description
par%spectral_type	'monochromatic'	Emission frequency distribution
par%temperature	10^4 K	Gas temperature
par%temperature0	$= T$	Source temperature for voigt0
par%bturb	0.0	Turbulent dispersion [km s^{-1}]
par%gaussian_sigma_vel	12.84	Gaussian σ [km s^{-1}] for spectral_type='gaussian'
par%gaussian_FWHM_vel	–999	Gaussian FWHM [km s^{-1}]; overrides σ if > 0
par%EW_line	0.0	Equivalent width [Å] for 'continuum+gaussian'
par%continuum_normalize	.true.	Normalize output so continuum level = 1
par%xfreq0	0.0	Frequency offset for emitted photons
par%line_prof_file	''	Custom line profile file
par%line_prof_file_type	0	0: (x, I); 1: (λ, I)
Frequency grid		
par%nxfreq	121	Spectral bins
par%xfreq_min, par%xfreq_max	auto	x-grid limits
par%nwavelength	0	Wavelength bins (overrides x-grid)
par%wavelength_min, par%wavelength_max	—	[Å]
par%nvelocity	0	Velocity bins (overrides x-grid)
par%velocity_min, par%velocity_max	—	[km s^{-1}]
Optical depth and density		
par%taumax	—	Line-center τ_0: center $\rightarrow +z$ boundary
par%tauhomo	—	Same, uniform-density equivalent
par%N_HImax	—	Column density: center $\rightarrow +z$ [cm^{-2}]
par%N_HIhomo	—	Same, uniform-density equivalent
par%N_gasmax	—	Ion column density [cm^{-2}] (metal lines with ion_model='none')

continued on next page

(continued)

Parameter	Default	Description
`par%density_rscale`	–999	Exponential density scale radius
`par%density_alpha`	0.0	Power-law density index: $n(r) \propto (r_{\max}/r)^{\alpha_D}$
External data		
`par%dens_file`	`''`	Density FITS cube [cm^{-3}]
`par%temp_file`	`''`	Temperature FITS cube [K]
`par%velo_file`	`''`	Velocity FITS cube [$\mathrm{km\,s}^{-1}$]
`par%input_field`	`''`	Base name for `.dens/.temp/.velo.fits.gz`
Velocity		
`par%velocity_type`	`''`	Velocity law (§6.4)
`par%Vexp`	0.0	Expansion/infall speed [$\mathrm{km\,s}^{-1}$]
`par%velocity_alpha`	1.0	Power-law velocity exponent for `'power_law'`
`par%Vx`, `par%Vy`, `par%Vz`	0.0	Uniform velocity [$\mathrm{km\,s}^{-1}$]
`par%Vpeak`	0.0	Peak velocity for SSH [$\mathrm{km\,s}^{-1}$]
`par%rpeak`	0.0	Radius at peak for SSH
`par%DeltaV`	0.0	Velocity increment for SSH [$\mathrm{km\,s}^{-1}$]
`par%Vrot`	0.0	Rotation speed [$\mathrm{km\,s}^{-1}$]
`par%recoil`	`.false.`	Photon recoil in scattering
AMR		
`par%amr_type`	`'ramses'`	`'generic'` or `'ramses'`
`par%amr_file`	`''`	AMR data path
`par%amr_snapnum`	1	RAMSES snapshot number
Clumpy medium		
`par%clump_radius`	—	Clump radius
`par%clump_tau0`	—	Line-center τ_0 per clump
`par%clump_f_cov`	—	Covering factor C_f
`par%clump_f_vol`	—	Volume filling factor f_V
`par%clump_N_clumps`	—	Number of clumps N

continued on next page

(continued)

Parameter	Default	Description
par%clump_sigma_v	0.0	Random velocity dispersion [km s^{-1}]
par%save_clump_info	.false.	Write clump table to FITS
Observers and peel-off		
par%nxim, par%nyim	0	Image pixels; peel-off active when > 0
par%dxim, par%dyim	auto	Pixel size [deg]
par%distance	$100\,r_{\max}$	Observer–center distance
par%obsx, par%obsy, par%obsz	$(0, 0, 1)$	Observer direction
par%alpha, par%beta	—	Azimuth, elevation [deg]
par%gamma	0.0	Detector-plane rotation [deg]
par%save_peeloff	.false.	1-D peel-off spectra
par%save_peeloff_2D	.false.	Peel-off spectral images
par%save_peeloff_3D	.false.	Full 3-D peel-off cube
par%save_direc0	.false.	Peel-off image without $e^{-\tau}$ (transparent-medium reference)
Output		
par%file_format	'hdf5'	Output container: 'hdf5' or 'fits'
par%out_file	''	Output file name (extension follows par%file_format)
par%out_merge	.false.	Accumulate into existing file
par%save_backup	.false.	Keep _NN copy of prior output on merge
par%out_bitpix	0	Storage precision: 0 = auto, −32, −64
par%save_Jin	.false.	Save injected spectrum
par%save_Jabs	.false.	Save absorbed spectrum
par%save_Jmu	.false.	Save angular spectrum $J(x;\mu)$
par%nmu	11	Number of μ bins for Jmu
par%save_all	.false.	Save full 3-D $J(\nu, x, y, z)$
par%save_radial_profile	.false.	Save 1-D radial J
par%save_sightline_tau	.false.	Save sightline τ

continued on next page

(continued)

Parameter	Default	Description
par%intensity_unit	''	Output header unit label
Dust		
par%DGR	0.0	Dust-to-gas ratio
par%albedo	0.32536	Single-scattering albedo
par%hgg	0.67617	Henyey–Greenstein asymmetry g
par%use_stokes	.false.	Polarization RT
par%scatt_mat_file	''	Mueller matrix file
par%use_reduced_wgt	.false.	Reduce weight instead of destroy
Parallelism		
par%use_master_slave	.true.	Dynamic load balancing
par%num_send_at_once	100	Photons per worker batch

15. Python Utilities

A small set of post-processing scripts is shipped under python/ in each version's source tree. They depend on numpy, astropy (for FITS inputs) and/or h5py (for HDF5 inputs), and (for plotting scripts) matplotlib. All scripts read both formats transparently via the lart_io.py backend (§15.2).

15.1 Loading and analysing output: read_lart.py

read_lart.py loads a LaRT spectral output (FITS or HDF5) into a rich LaRTOutput dataclass that aggregates the angle-integrated spectrum, the optional $J(\mu,x)$ output, and every peel-off observer file alongside it. The dataclass exposes a small library of plotting and analysis methods so common diagnostics can be produced in a few lines. The script accepts the input-file name in any of four forms:

- full input file: t1tau2.in
- stem only: t1tau2 (.in is added automatically)
- the HDF5 file: t1tau2.h5 (also .hdf5)
- the FITS file: t1tau2.fits.gz (also .fits)

When given an input file, the reader honors par%out_file from the namelist first, then searches for any LaRT output stem under all recognised extensions (HDF5 preferred) via lart_io.find_lart_file. If the output file is still absent but the namelist sets par%clump_input_file pointing at an existing clump file, read_lart prints a one-line notice and returns a "clumps-only" LaRTOutput with the spectrum arrays unset (out.is_clumps_only == True) but with the clump population already loaded into out.clumps (a ClumpsOutput, see below). out.plot_clump_slice() therefore

works on the pre-run dataclass, so the clump population can be inspected before launching the radiative-transfer run. Peel-off observer files are auto-discovered the same way — both the single-observer (`<stem>_obs.h5`) and the multi-observer (`<stem>_obs_000.h5`, `_obs_001.h5`, ...) layouts are supported, in either format.

Module API.

```
from read_lart import read_lart
out = read_lart('t1tau2.in')      # FITS or HDF5 -- format-agnostic

# 1-D arrays, length nxfreq
out.xfreq, out.velocity, out.wavelength
out.Jout                  # always present
out.Jin, out.Jabs, out.Jabs2     # optional -- None if not in file

# Jmu output -- present only if par%save_Jmu = .true. was used
out.mu          # bin centers, length nmu
out.mu_edges    # bin edges, length nmu+1
out.Jmu         # shape (nmu, nxfreq); None if absent
out.nmu, out.mu_min, out.dmu

# Peel-off data for each observer (one PeelObservation per observer file)
out.peelings    # list, sorted by (beta, alpha, gamma)

# Raw section attribute dicts (case-insensitive; legacy uppercase
# keys like spec_hdr['EXTNAME'] also work)
out.spectrum_header, out.jmu_header
```

Every optional array is `None` when its section is not present, so the reader works on outputs from any LaRT version and either format. The `spectrum_header` and `jmu_header` dicts are filled with both the original-case keys (as they appear in the underlying file) and uppercase aliases, so legacy callers using FITS-style key access (e.g. `spec_hdr['EXTNAME']`) keep working on HDF5 outputs.

Peel-off observer accessor. Each entry in `out.peelings` is a `PeelObservation` dataclass with fields `alpha`, `beta`, `gamma`, `distance`, `nphotons`, `nxim`, `nyim`, `sr_pix`, the scattered + direct cubes `scatt`, `direc`, `direc0` (each shape `(nyim, nxim, nxfreq)`), and the combined cube `= scatt + direc`. Two helper methods are provided:

```
po = out.peelings[0]
spec = po.average_spectrum()       # area-mean spectrum, length nxfreq
v0, vsig = po.velocity_moment_map(weight='cube')   # (nyim, nxim) each
```

Plotting Jmu. `LaRTOutput.plot_jmu()` draws the angular spectrum directly:

```
out = read_lart('t1tau3.in')
out.plot_jmu(kind='lines', x='velocity')   # one curve per mu bin
out.plot_jmu(kind='image', x='velocity')   # 2-D heatmap (x vs mu)
```

The x= keyword selects the abscissa (`'velocity'`, `'xfreq'`, or `'wavelength'`); the `kind=` keyword toggles between line ensemble (colored by μ) and a 2-D image. The line view also overplots `Jout` as a black dashed reference. Pass `ax=` to draw onto an existing axes, `savefig='out.png'` to save to disk. Axis limits accept either the tuple form `xlim=(lo,hi)` / `ylim=(lo,hi)` or the shorthand `xmin/xmax/ymin/ymax` for each bound; the individual-bound override takes precedence when both are given:

```
out.plot_jmu(kind='lines', xlim=(-500, 500), ymin=0)
out.plot_jmu(kind='lines', xmax=300)        # left edge auto-scaled
```

CALCJ / CALCP profiles. When the output was produced by a `CALCJ` / `CALCP` / `CALCPnew` build (§10.3), the sections are loaded automatically: `out.J1` (shape `(nbin, nxfreq)`), `out.P1`, `out.P1_new`, the 2-D / 3-D, and leaf-indexed variants (`J2`, `J3D`, `J_leaf`, ...), and the bin axes `out.r_JPa` / `out.z_JPa` (reconstructed from the section keywords for AMR runs, or from the namelist for Cartesian runs). Two plot helpers draw the 1-D profiles:

```
out.plot_J_profile()                    # freq-integrated <J>(r) or <J>(z)
out.plot_Pa_profile(which='auto')       # P_alpha(r); 'Pa' or 'Pa_new'
```

Peel-off analysis plots. Several methods operate on the auto-discovered `out.peelings` list and produce moment maps, spectra for each observer, and radial profiles:

```
out.plot_peeling_map(weight='cube',     mode='intensity')
out.plot_peeling_map(weight='velocity', mode='v_los')
out.plot_peeling_map(weight='velocity', mode='sigma_v')
out.plot_peeling_spectrum()              # one curve per observer
out.plot_peeling_radial_profile()        # azimuthally averaged
out.plot_velocity_moment_map(po, mode='v_los')   # single observer
```

The `plot_peel_jmu_compare()` method overlays each observer's peel-off spectrum on the matching $J(\mu,x)$ slice as an isotropy diagnostic, and `plot_clump_slice()` draws the 2D cross-section of the clumpy medium at a chosen plane (it opens the matching `<stem>_clumps.h5` / `.fits.gz` file automatically). All plotting methods accept `ax=`, `savefig=`, and `xlim/ylim` (or `xmin/xmax/ymin/ymax`) the same way as `plot_jmu`.

Command-line usage.

```
python read_lart.py t1tau2          # auto-resolves to t1tau2.in
python read_lart.py t1tau2.in
python read_lart.py t1tau2.h5       # HDF5 output
python read_lart.py t1tau2.fits.gz  # FITS output
```

prints a one-screen summary of which arrays are present and, when `Jmu` is in the file, the $\max_x J(x;\mu)\,/\,\max_x J_{\rm out}(x)$ ratio in each bin (a quick isotropy sanity check for homogeneous models).

Standalone clump access: `read_clumps` / `read_clump`. The clump population can also be loaded independently of any radiative- transfer output via the top-level helper:

```
from read_lart import read_clumps      # alias: read_clump
```

```
co = read_clumps('foo_clumps.h5')      # direct clump file
co = read_clumps('run.in')             # via par%clump_input_file
co = read_clumps('run')                # stem -> tries .in, then _clumps.*
print(co.summary())
co.plot_clump_slice(axis='z')
```

The returned `ClumpsOutput` dataclass exposes the clump arrays (`x, y, z, vx, vy, vz, radius, rhokap, density, temp`, plus the convenience attributes `pos`, `vel`, `n_clumps`, `sphere_r`, `f_vol`, `f_cov`, `distance_unit`, `distance2cm`), the case-insensitive attribute dict `attrs` (with `co.attr(name)` helper), and the `plot_clump_slice()` method. The opacity column in the file is auto-detected with priority `RHOKAP` → `DENSITY` → `DENS`: `RHOKAP` populates `co.rhokap` (opacity per code unit), while `DENSITY` / `DENS` populate `co.density` (n_{HI} [cm^{-3}]); `co.opacity_col` records which column was found. `co.distance_unit` / `co.distance2cm` expose the `DISTUNIT` / `DIST_CM` attributes from the file (LaRT's `par%distance_unit` / `par%distance2cm` at write time). When the `F_VOL` / `F_COV` attributes are missing from the file, the `f_vol` / `f_cov` properties fall back to `compute_f_vol()` / `compute_f_cov()`, which evaluate the same formulas as LaRT's `write_clumps_fits()` from-file branch in `clump_mod.f90`: $f_{\rm vol} = \sum r_i^3/(R^3 - r_{\rm min}^3)$ and $f_{\rm cov} = \frac{3}{4}\sum r_i^2/(R^2 + R\,r_{\rm min} + r_{\rm min}^2)$, with R taken from `SPHERE_R` (or `RMAX`) and $r_{\rm min}$ from `RMIN` (default 0). The two `compute_*` methods can also be called explicitly to recompute regardless of the stored attribute. Inside `LaRTOutput` the same dataclass lives at `out.clumps` (lazy- loaded on the first `plot_clump_slice` call, or populated eagerly in the clumps-only pre-run state described above).

Other scripts. The following plotting helpers are also shipped:

- `plot_spectrum.py`, `plot_map.py`, `plot_intensity_profile.py`, `plot_Pa.py`: standard quick-look plots.
- `check_flux.py`, `flux_check.py`: flux-conservation diagnostics.
- `AMR_grid/`: helpers for generating generic AMR data files (see `AMR_grid/make_amr_grid.py`, `make_amr_sphere_radial.py`). For velocity fields, `make_amr_grid.py --vel_criterion doppler --vel_threshold 1.0` refines each leaf until the bulk velocity changes across it by at most one Doppler width $b = \sqrt{2kT/m + b_{\rm turb}^2}$.

15.2 Format-agnostic reader and FITS ↔ HDF5 converter: `lart_io.py`

`python/lart_io.py` provides a single `load_lart(path)` entry point that reads either a FITS or an HDF5 LaRT output and returns a uniform `LartFile` structure (one `Section` per HDU / group, with `.data` or `.columns` plus `.attrs`). It also converts between the two formats by extension. Requires `numpy` plus `astropy` (for FITS) and/or `h5py` (for HDF5); each backend is only imported when needed.

Module API.

```
from lart_io import load_lart, convert

f = load_lart('run.h5')            # works on .fits.gz too
spec = f['Spectrum']               # a Section
```

```
jout = spec.columns['Jout']       # 1-D numpy array
nph  = spec.attrs['nphotons']

scat = load_lart('run_obs.h5')['Scattered']
cube = scat.data                  # 3-D array (nxim, nyim, nxfreq)

convert('run.fits.gz', 'run.h5')  # both directions by extension
convert('run.h5',      'run.fits.gz')
```

The round-trip preserves all numerical data bit-identically; string keyword names are normalized to uppercase on the FITS leg (FITS card-name convention), which is the only visible difference between the two forms.

Command-line usage.

```
python lart_io.py info  run.h5
python lart_io.py info  run.fits.gz
python lart_io.py convert  run.fits.gz  run.h5
python lart_io.py convert  run.h5       run.fits.gz
```

16. Example Input Files

16.1 Uniform Sphere, Lyα, Static

```
&parameters
 par%no_photons     = 1e6
 par%temperature    = 1.0e4
 par%taumax         = 1.0e4
 par%rmax           = 1.0
 par%nx = 101, par%ny = 101, par%nz = 101
 par%source_geometry = 'point'
 par%spectral_type  = 'voigt'
 par%nxfreq         = 121
/
```

16.2 Sphere with Hubble Outflow and Peel-off

```
&parameters
 par%no_photons     = 1e6
 par%temperature    = 1.0e4
 par%taumax         = 1.0e4
 par%rmax           = 1.0
 par%nx = 101, par%ny = 101, par%nz = 101
```

```
 par%source_geometry = 'point'
 par%spectral_type  = 'voigt'
 par%velocity_type  = 'hubble'
 par%Vexp           = 200.0          ! km/s outflow
 par%nxfreq         = 121
 par%nxim = 129, par%nyim = 129
 par%alpha = 0.0  45.0  90.0
 par%beta  = 0.0   0.0   0.0
 par%save_peeloff_2D = .true.
/
```

16.3 Realistic Galaxy (FITS Density Field, Stokes Polarization)

```
&parameters
 par%no_photons    = 1e8
 par%dens_file     = 'galaxy.dens.fits.gz'
 par%temp_file     = 'galaxy.temp.fits.gz'
 par%velo_file     = 'galaxy.velo.fits.gz'
 par%taumax        = 1.0e4           ! rescale to this optical depth
 par%distance_unit = 'kpc'
 par%source_geometry = 'exponential'
 par%source_rscale = 3.0
 par%source_zscale = 0.5
 par%spectral_type = 'voigt'
 par%DGR           = 1.0
 par%use_stokes    = .true.
 par%nxfreq        = 121
 par%nxim = 200, par%nyim = 200
 par%save_peeloff_2D = .true.
/
```

16.4 AMR Sphere (Generic Text Format)

```
&parameters
 par%use_amr_grid   = .true.         ! or par%grid_type = 'amr' in v2.10
 par%amr_type       = 'generic'
 par%amr_file       = 'sphere_amr.dat'
 par%no_photons     = 1e6
 par%taumax         = 1.0e4
 par%source_geometry = 'point'
```

```
 par%spectral_type  = 'voigt'
 par%nxfreq         = 121
 par%nxim = 100, par%nyim = 100
 par%save_peeloff_2D = .true.
 par%save_Jin       = .true.
/
```

16.5 Clumpy Sphere with Random Velocities

```
&parameters
 par%use_clump_medium = .true.       ! or par%grid_type = 'clump' in v2.10
 par%rmax             = 1.0
 par%clump_radius     = 0.01
 par%clump_f_cov      = 5.0
 par%clump_tau0       = 1.0e3
 par%temperature      = 1.0e4
 par%clump_sigma_v    = 50.0         ! km/s random dispersion
 par%no_photons       = 1e6
 par%spectral_type    = 'voigt'
 par%nxfreq           = 301
 par%velocity_min = -200.0
 par%velocity_max =  200.0
 par%save_clump_info  = .true.
/
```

16.6 Illustris/TNG Galaxy (Diffuse Emissivity)

After converting a TNG cutout with `convert_illustris_to_generic.py --compute-physics`:

```
&parameters
 par%use_amr_grid       = .true.
 par%amr_type           = 'generic'
 par%amr_file           = 'galaxy.h5'
 par%distance_unit      = 'kpc'
 par%no_photons         = 1e6
 par%spectral_type      = 'voigt'
 par%source_geometry    = 'diffuse_emissivity'
 par%core_skip          = .true.
 par%nxfreq             = 201
 par%velocity_min       = -1000.0
 par%velocity_max       =  1000.0
```

```
 par%save_peeloff       = .true.
 par%nxim = 128, par%nyim = 128
 par%distance           = 1e4
 par%alpha = -90.0
 par%beta  =   0.0
/
```

The `emissivity` column in `galaxy.h5` is automatically used as the photon source distribution. No `par%emiss_file` or `par%emissivity_model` is required.

Metal-line Examples (C IV, O VI)

The converter supports CIE collisional excitation emissivity for metal lines via `--emissivity-line`:

```
# C IV 1548+1550
python convert_illustris_to_generic.py cutout.hdf5 \
    -o galaxy_civ.h5 --compute-physics --emissivity-line civ

# O VI 1032+1038
python convert_illustris_to_generic.py cutout.hdf5 \
    -o galaxy_ovi.h5 --compute-physics --emissivity-line ovi
```

Then run LaRT with the appropriate `par%line_id`:

```
&parameters
 par%use_amr_grid       = .true.
 par%amr_type           = 'generic'
 par%amr_file           = 'galaxy_ovi.h5'
 par%distance_unit      = 'kpc'
 par%line_id            = 'OVI_1032'
 par%no_photons         = 1e6
 par%spectral_type      = 'voigt'
 par%source_geometry    = 'diffuse_emissivity'
 par%core_skip          = .true.
 par%nxfreq             = 201
 par%velocity_min       = -1500.0
 par%velocity_max       =  2500.0
/
```

Note: the O VI doublet separation is $\sim 1652\,\mathrm{km\,s^{-1}}$ (vs. $\sim 500\,\mathrm{km\,s^{-1}}$ for C IV), so a wider velocity range is needed.